Feedback amplifiers

Why?

- Basic amplifier is not stable because of perturbations due to temperature and aging effects
- Distortions at output
- Input and output impedance is not as per our requirement/desire

Components of Basic amplifier

- Basic amplifier
- Feedback Network
- Mixer (Adder or Subtractor

Feedback network

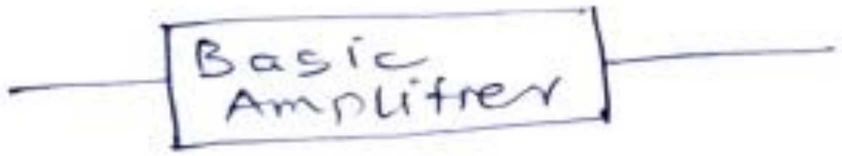
- Resistive circuit
- Small transfer function
- Phase angle-may be present or not present

Feedback network – Transfer function

Mixer_ In phase/ out of phase

Effect of Feedback on Cutoff frequency & Bandwidth of amplifier

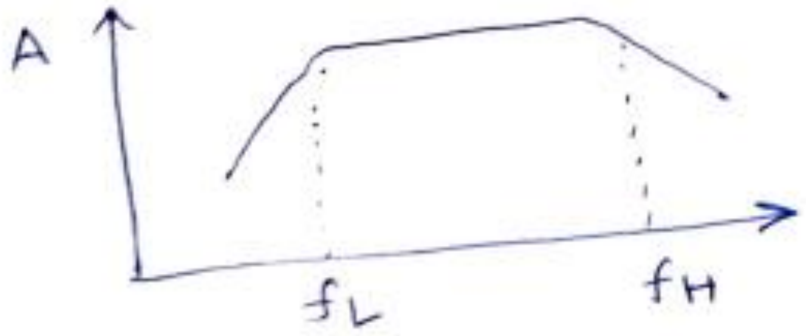
β = Transfer Ratio

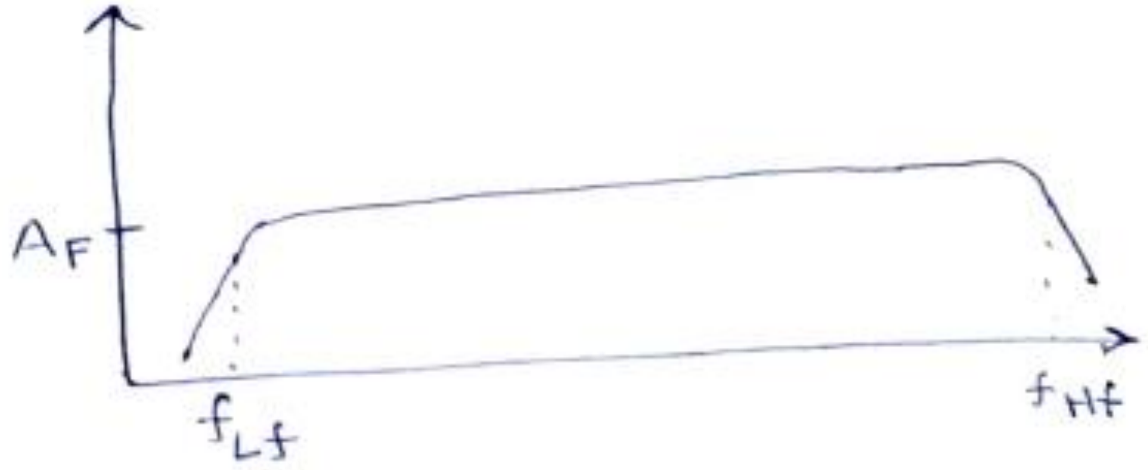


f_L, f_H = Lower & Higher cutoff frequencies

f_{Lf}, f_{Hf} = Lower & Higher cut off frequencies

$$f_{Lf} = \frac{f_L}{1+A\beta} \; ; \; f_{HF} = f_H(1+A\beta)$$





$A_f << A$

$BW_f > BW$

Gain Bandwidth remain fixed

$$A \times BW = A_f \times BW_f$$

Types of Amplifiers

Amplifier Types

Nature of I/P signal

- Voltage waveform
- Current waveforms

Nature of O/P signal

- Voltage waveform
- Current waveform

① Voltage amplifier

I/P - Voltage waveform O/P - Voltage waveform

② Current Amplifier

I/P - ~~Voltage~~ current waveform O/P - current waveform

③ Transconductance Amplifiers

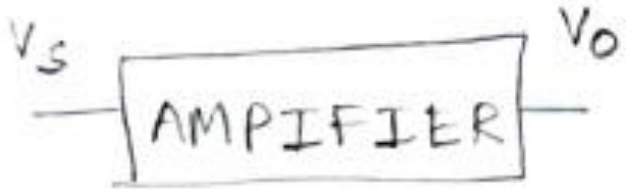
I/P - Voltage w/f O/P - current w/f

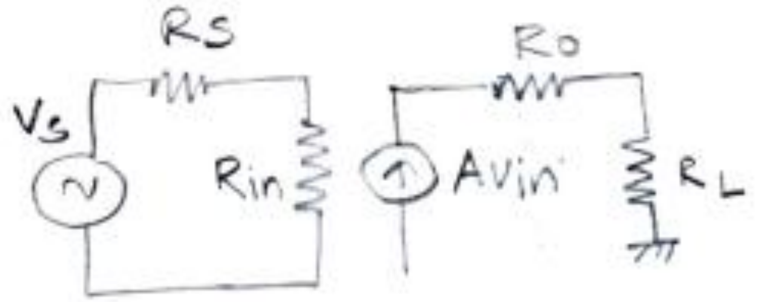
V → I converter

④ Transresistance Amplifiers

I/P - current w/f O/P - voltage W/f

Voltage Amplifier





So,

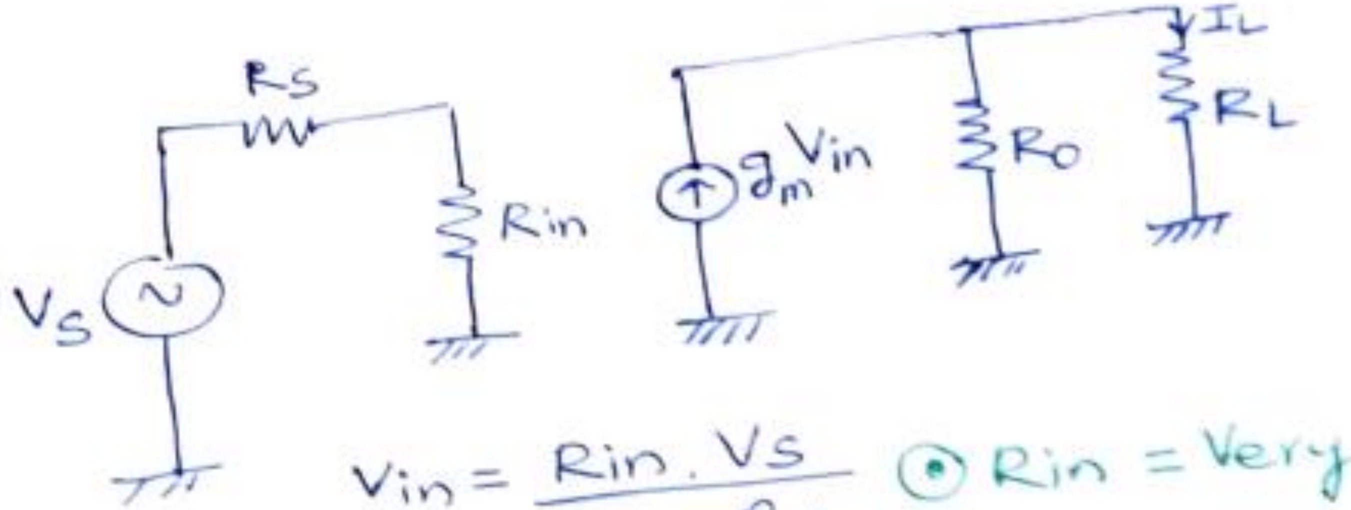
$R_{in} \rightarrow$ Very high

$R_o \rightarrow$ Very Low

$$V_{in} = \frac{R_{in}}{R_{in}+R_s} V_s$$

$$V_{out} = \frac{R_L}{R_o+R_L} AV_{in}$$

TRANSCONDUCTANCE AMPLIFIER



$$V_{in} = \frac{R_{in} \cdot V_s}{R_{in}+R_s}$$

⊙ R_{in} = Very high

⊙ R_o = Very high

$$I_L = \frac{R_o}{R_o+R_L} g_m V_{in}$$

$V_{in} \simeq V_s$

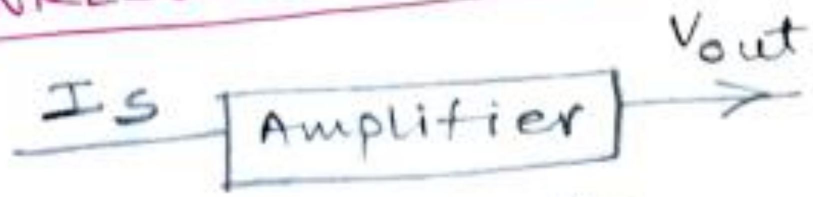
Current Amplifier

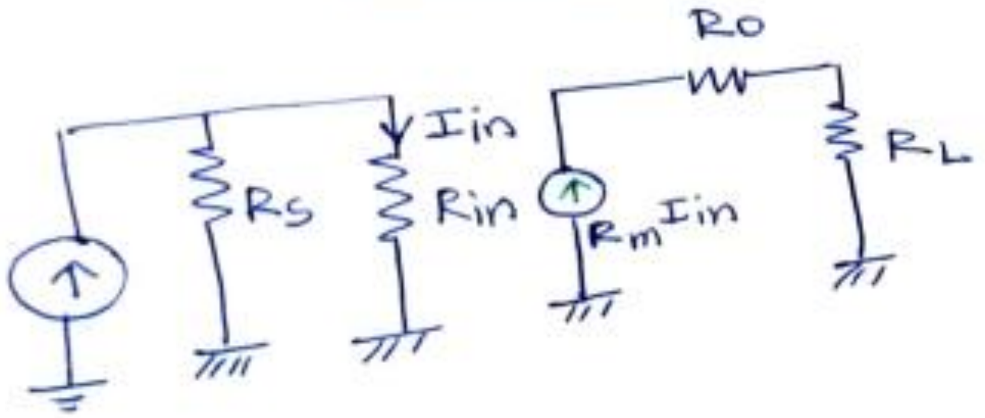
current source has high internal resistance

⊙ R_{in} = Very small

⊙ R_o = Very Large

TRANRESISTANCE AMPLIFIER





$$I_{in} = \frac{R_s I_s}{R_s+R_{in}}$$

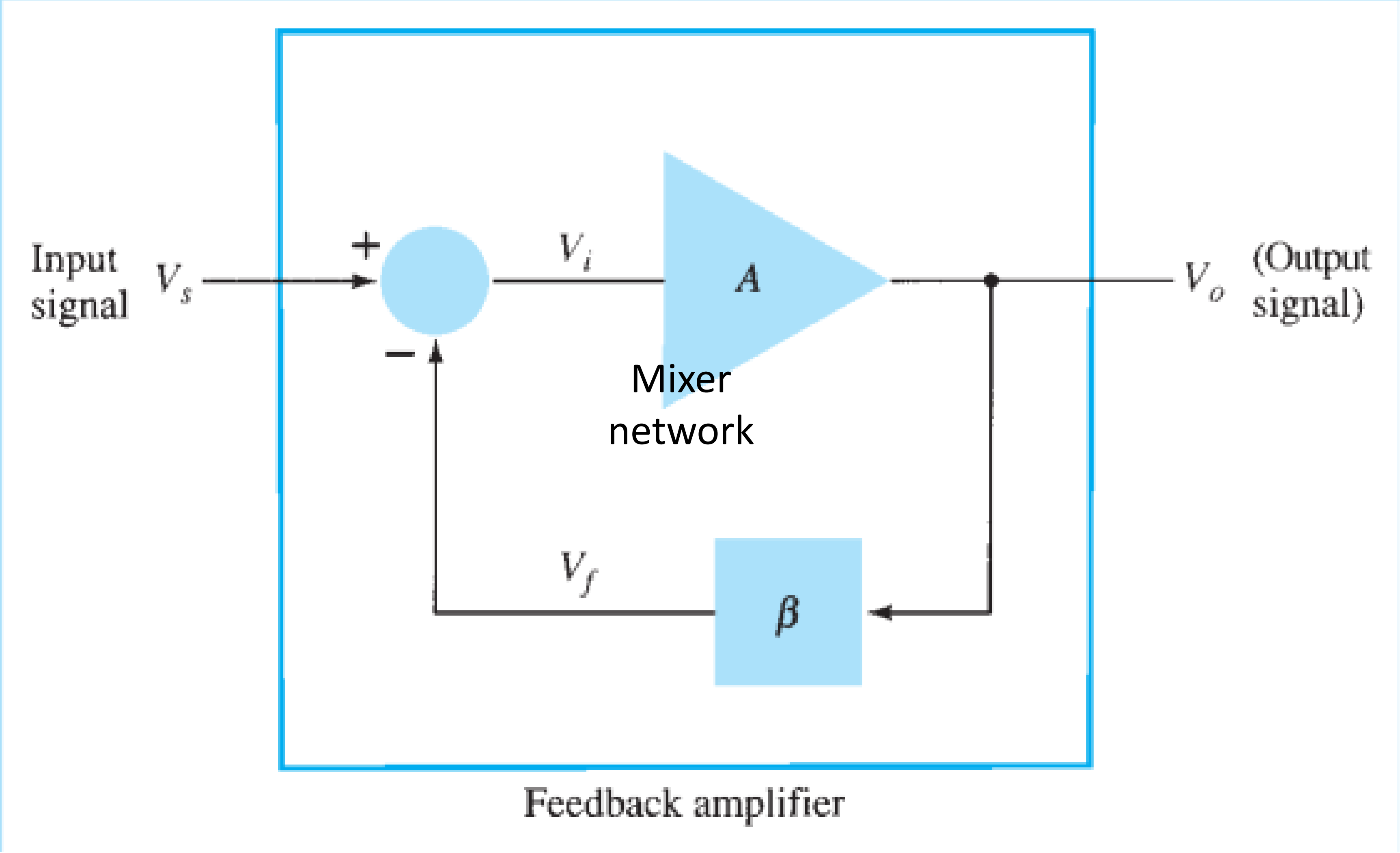
⊙ R_{in} = very small.

⊙ R_o = Very Low

$I_{in} \simeq I_s$

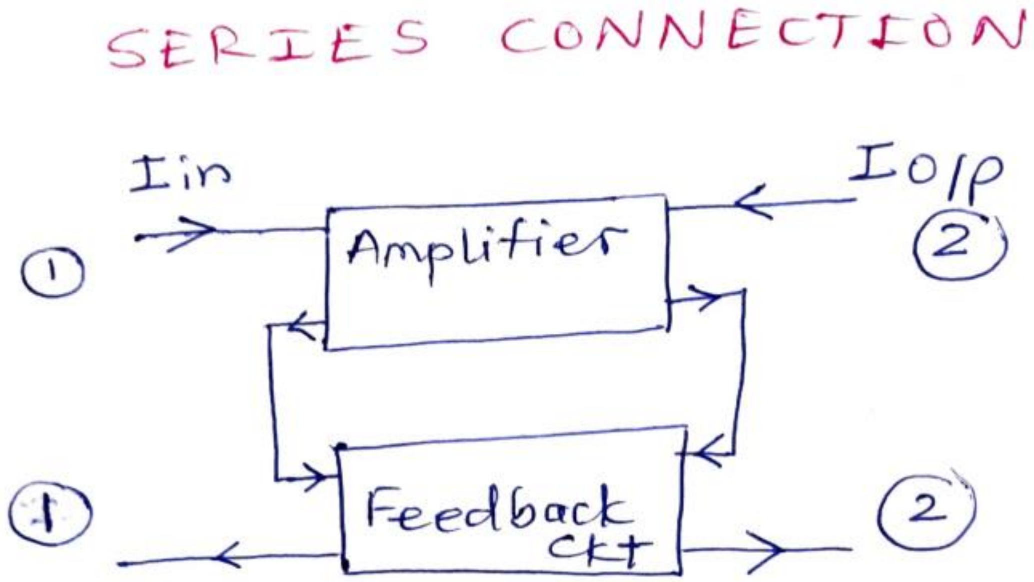
$$V_o = \frac{R_L}{R_L+R_o} \approx R'_m I'_{in} \simeq R_m I_{in}$$

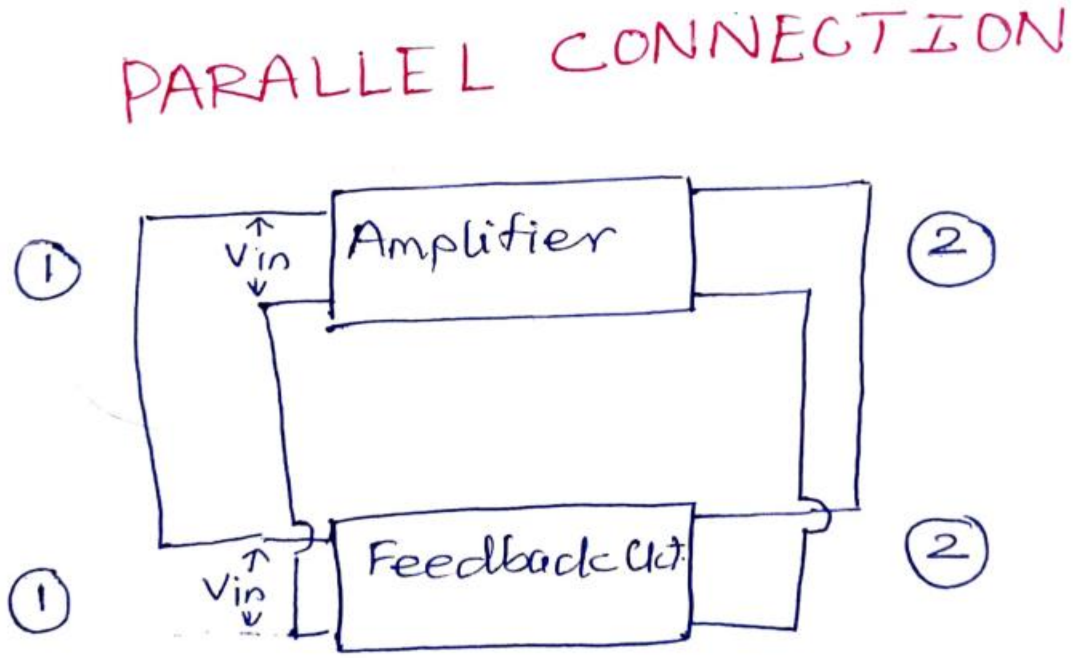
Feedback in amplifiers



A typical block diagram of feedback amplifier

- The input signal Vs is applied to a mixer network, where it is combined with a feedback signal Vf .
- The difference of these signals Vi is then the input voltage to the amplifier.
- A portion of the amplifier output Vo is connected to the feedback network (b), which provides a reduced portion of the output as feedback signal to the input mixer network.





Feedback Amplifier

Advantages

If the feedback signal is of opposite polarity to the input signal, negative feedback results. Although negative feedback results in reduced overall voltage gain, a number of improvements are obtained:

1. Optimized input impedance.
2. Better stabilized voltage gain.
3. Improved frequency response.
4. Optimized output impedance.
5. Reduced noise.
6. More linear operation.

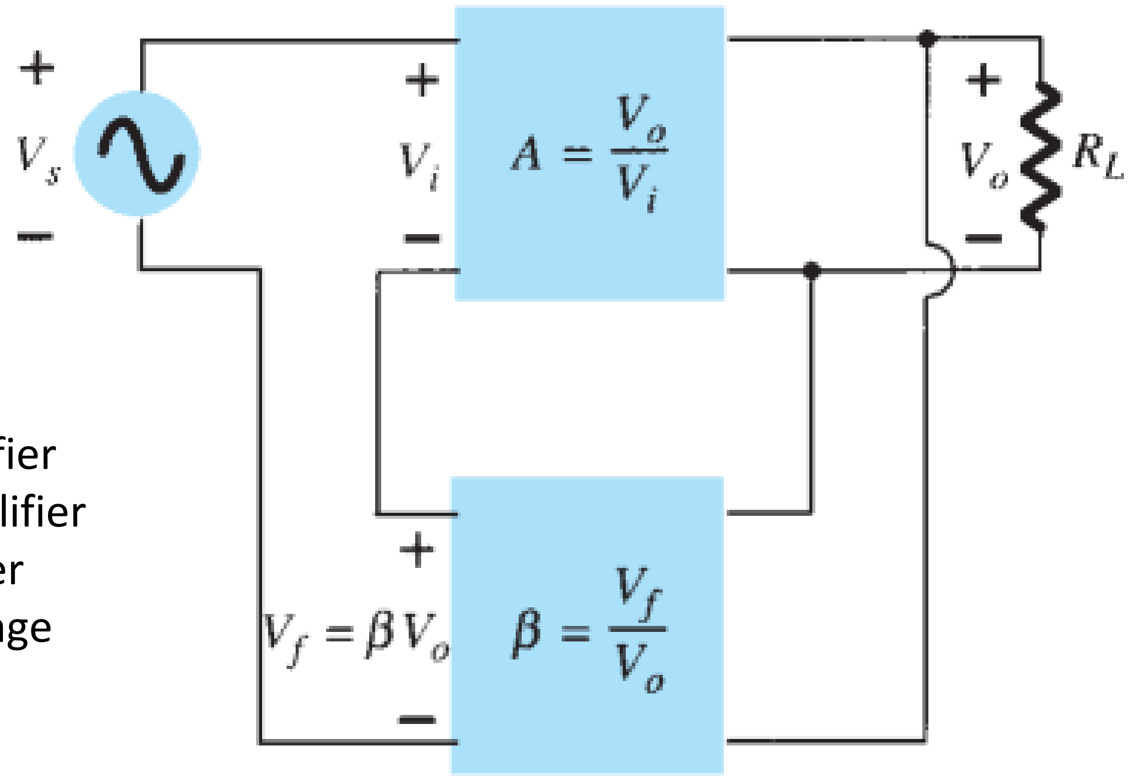
Feedback types

There are four basic ways of connecting the feedback signal. Both voltage and current can be fed back to the input either in series or parallel.

1. Voltage-series feedback
2. Voltage-shunt feedback

3. Current-series feedback
4. Current-shunt feedback

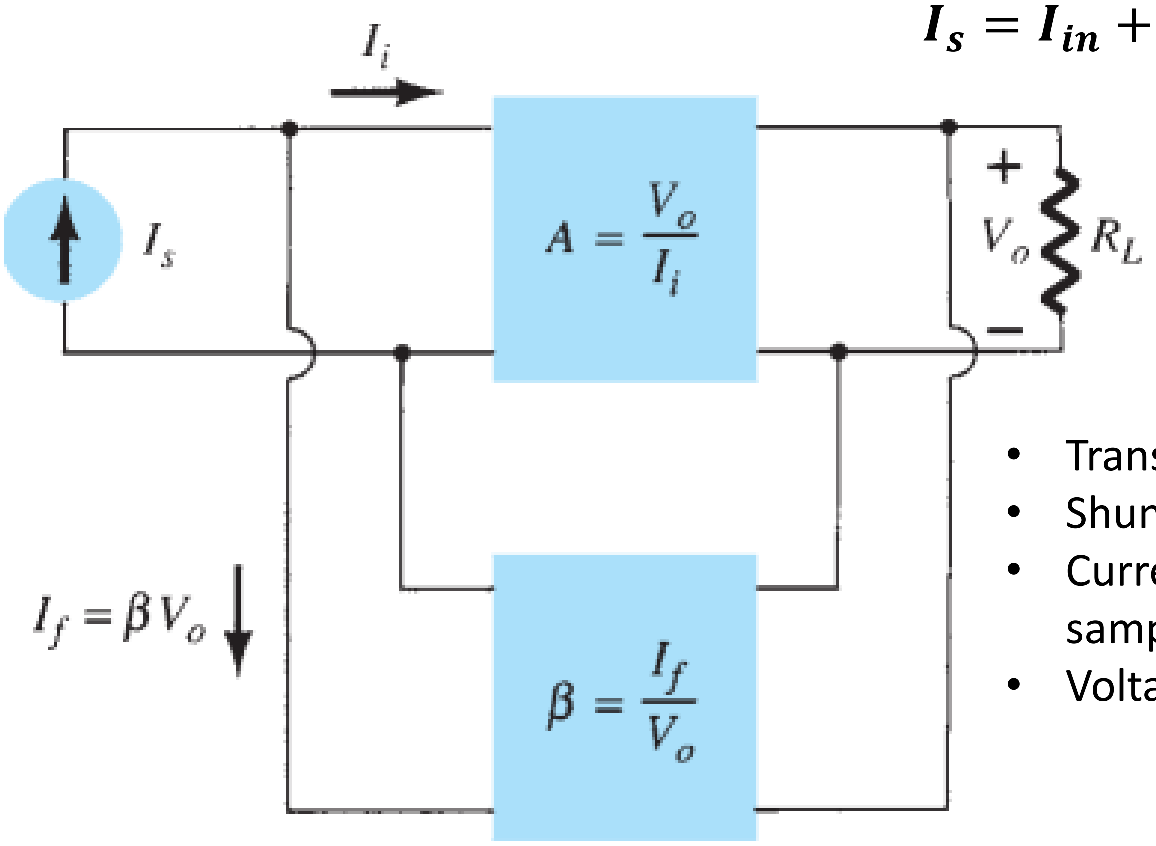
$$V_s = V_{in} + V_f$$



- Voltage series amplifier
- Voltage voltage amplifier
- Series shunt amplifier
- Voltage mixing, voltage sampling amplifier

1. Voltage-series feedback ($A_f = V_o/V_s$)

$$I_s = I_{in} + I_f$$



- Trans-resistance amplifier
- Shunt shunt amplifier
- Current mixing, voltage sampling amplifier
- Voltage current amplifier

2. Voltage-shunt feedback ($A_f = V_o/I_s$)

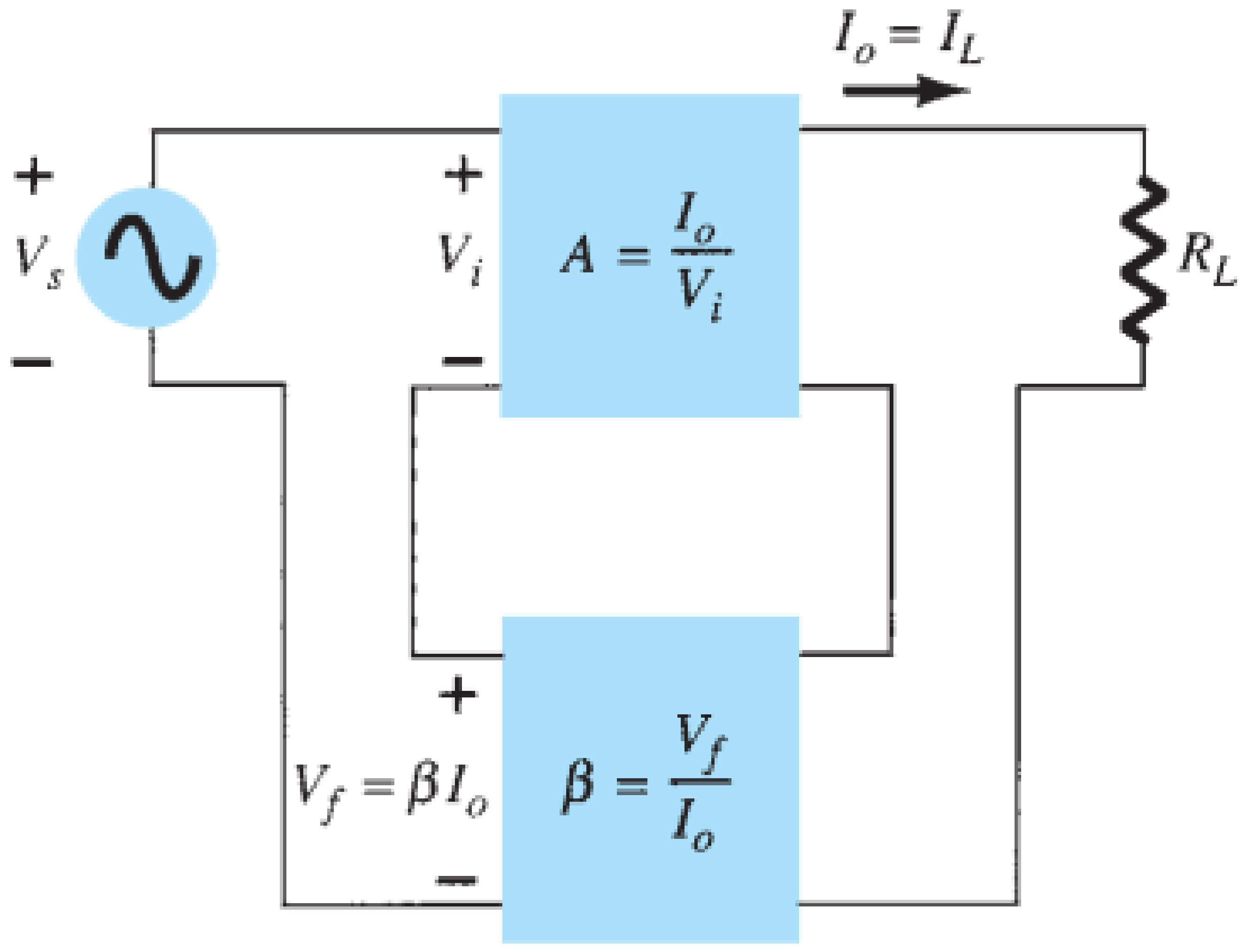
Nomenclature

- Connection Based
- Sampling – mixing connection
- Sampling mixing parameter

Here,

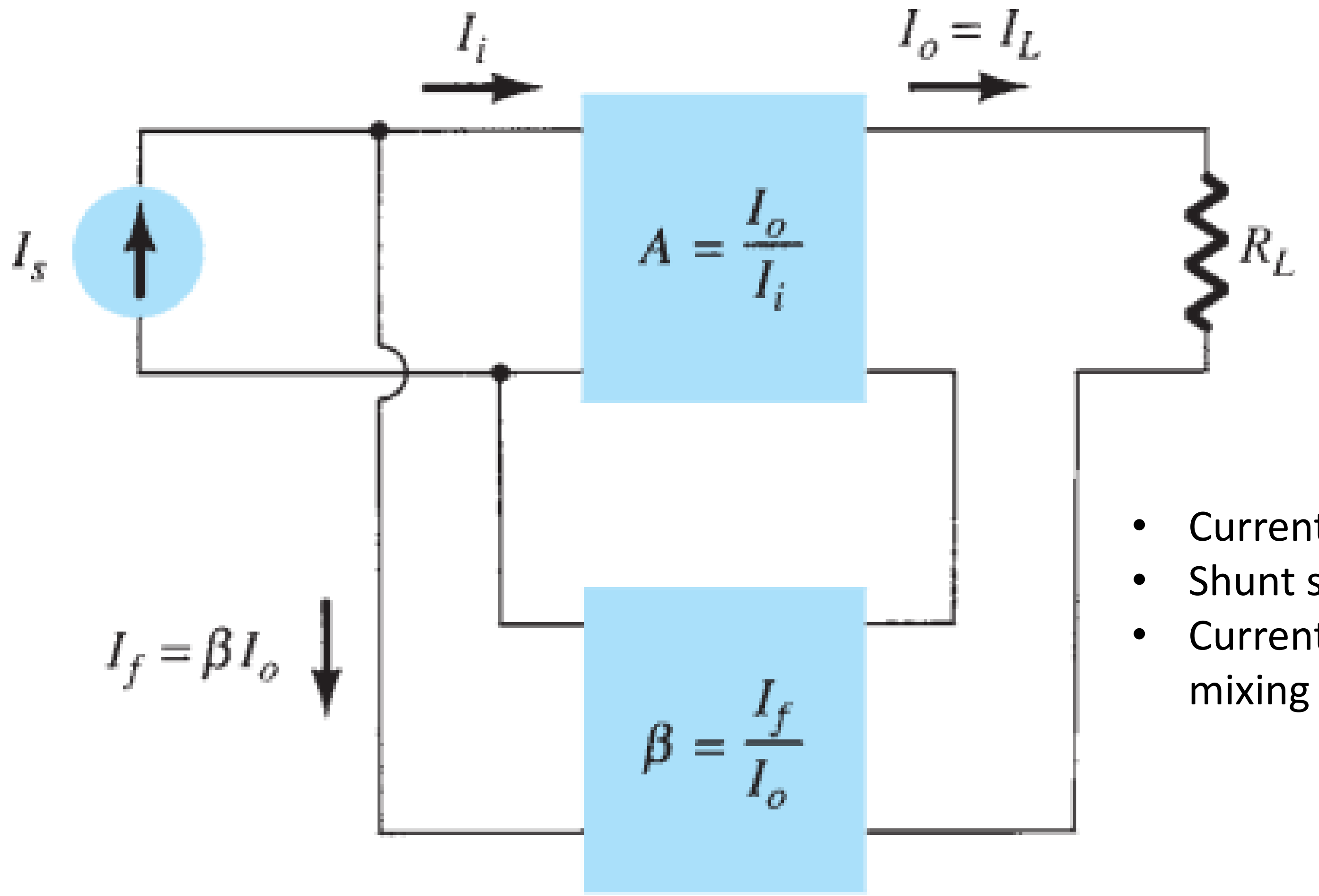
- voltage refers to connecting the output voltage as input to the feedback network
- current refers to tapping off some output current through the feedback network.
- Series refers to connecting the feedback signal in series with the input signal voltage;
- shunt refers to connecting the feedback signal in shunt (parallel) with an input current source.

Feedback types



3. Current-series feedback ($A_f = I_o/V_s$)

Series series amplifier
Current sampling, voltage mixing amplifier
Current voltage amplifier



- Current shunt amplifier
- Shunt series amplifier
- Current sampling, voltage mixing amplifier

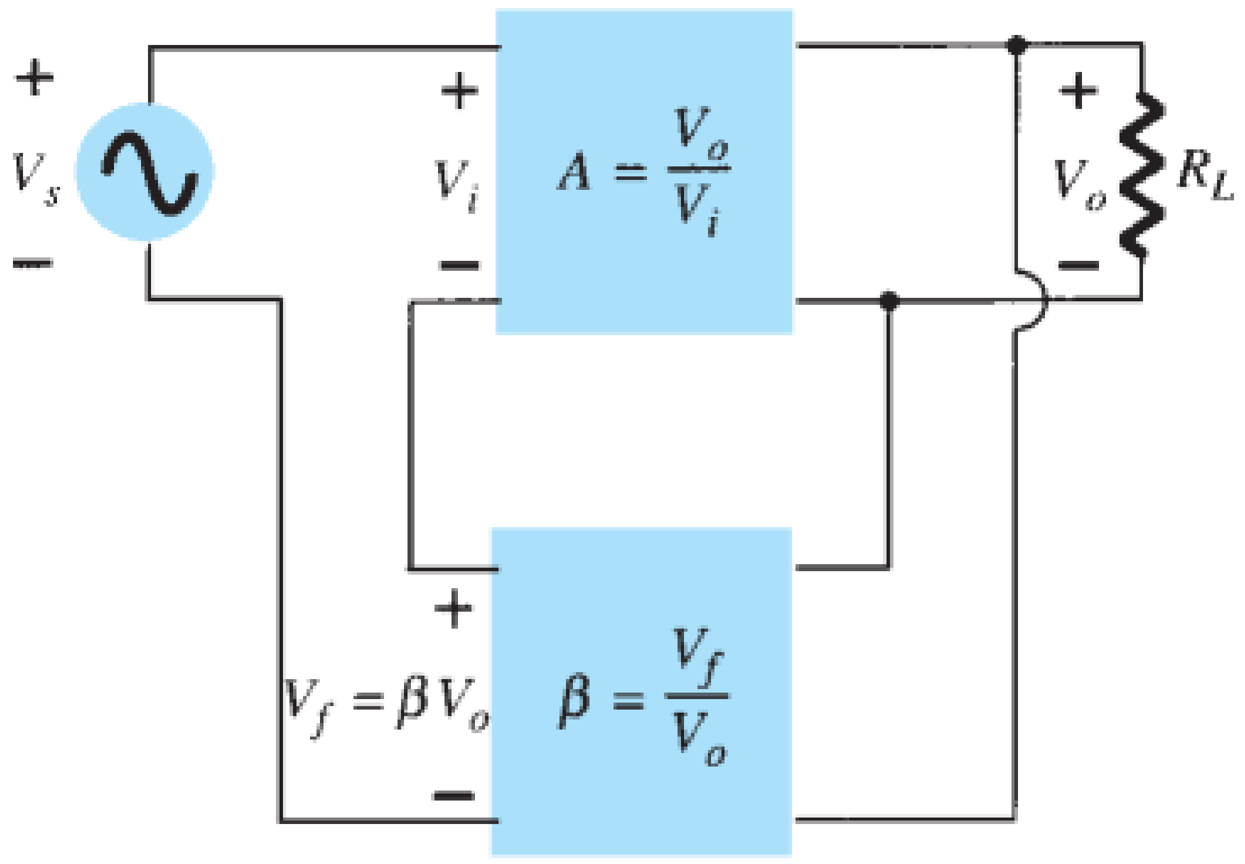
4. Current-shunt feedback ($A_f = I_o/I_s$)

Here,

- current refers to tapping off some output current through the feedback network.
- Series refers to connecting the feedback signal in series with the input signal voltage;
- shunt refers to connecting the feedback signal in shunt (parallel) with an input current source.

Gain with feedback

Voltage series feedback amplifier



The voltage-series feedback connection with a part of the output voltage fed back in series with the input signal, resulting in an overall gain reduction. If there is no feedback (Vf = 0), the voltage gain of the amplifier stage is

If there is no feedback $A = \frac{V_o}{V_s} = \frac{V_o}{V_i}$

If a feedback signal V_f is connected in series with the input, then

$$V_i = V_s - V_f$$

Since $V_o = AV_i = A(V_s - V_f) = AV_s - AV_f = AV_s - A(\beta V_o)$

then $(1 + \beta A)V_o = AV_s$

so that the overall voltage gain *with* feedback is

$$\boxed{A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A}}$$

So, the overall gain of the circuit is reduced by a factor (1+ βA)

$$A_f = \frac{V_o}{V_f} = \frac{A}{1+\beta A}$$

$A >> 1$; $AB >> 1$

$1 + AB \simeq AB$

So $\boxed{A_f = \frac{1}{B}}$

Amount of feedback : It is difference of close loop gain and open loop gain in dB.

Amount of feedback = $20\log(A_f) - 20\log(A)$

$20\log\frac{A_f}{A} = 20\log\left(\frac{1}{1+BA}\right)$

$-20\log(1+\beta A)$

Gain with feedback

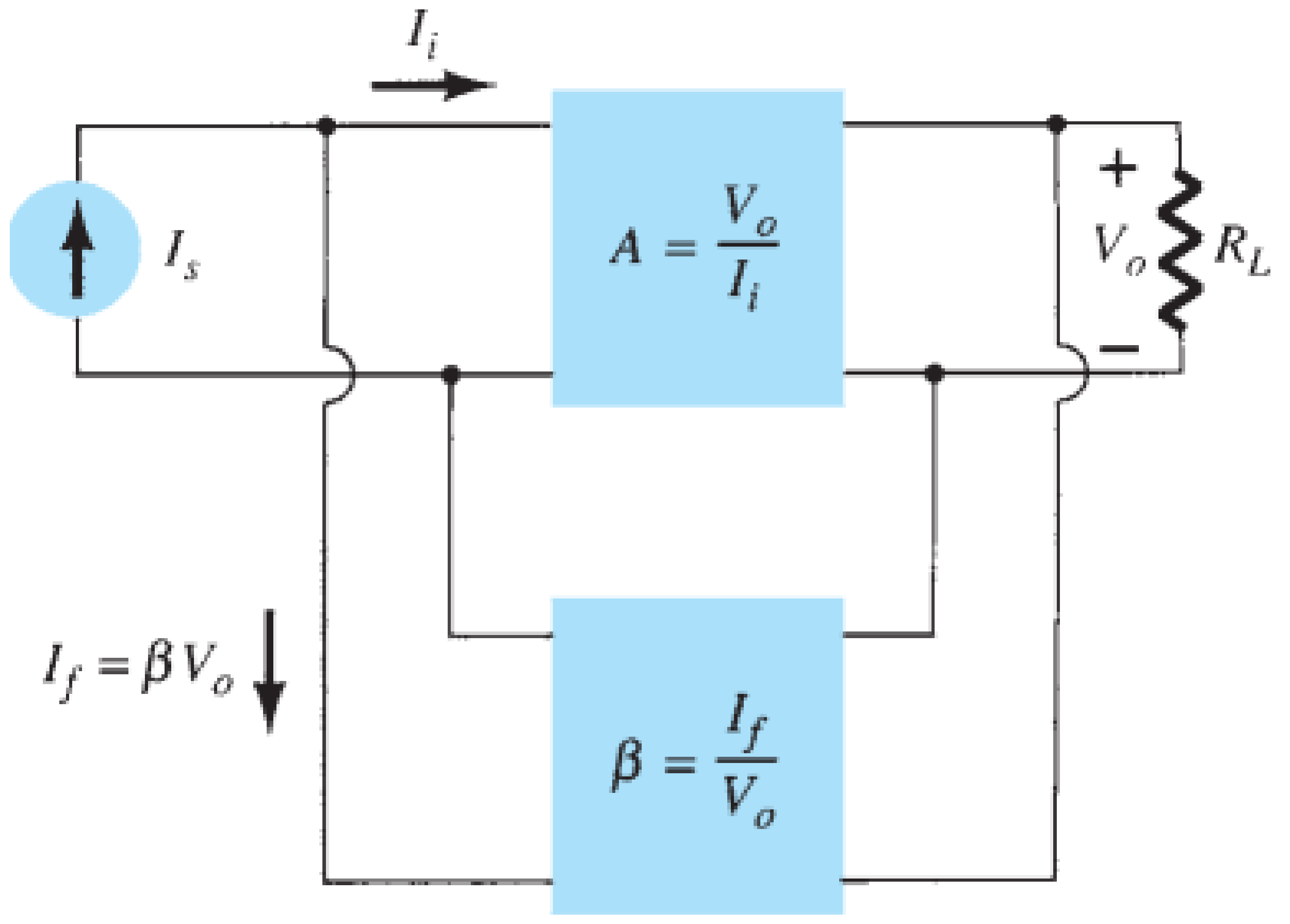
Voltage shunt feedback amplifier

Gain with feedback for the amplifier is given by

$$A_f = \frac{V_o}{I_s} = \frac{A\,I_i}{I_i + I_f} = \frac{A\,I_i}{I_i + \beta V_o} = \frac{A\,I_i}{I_i + \beta A\,I_i}$$

$$\boxed{A_f = \frac{A}{1 + \beta A}}$$

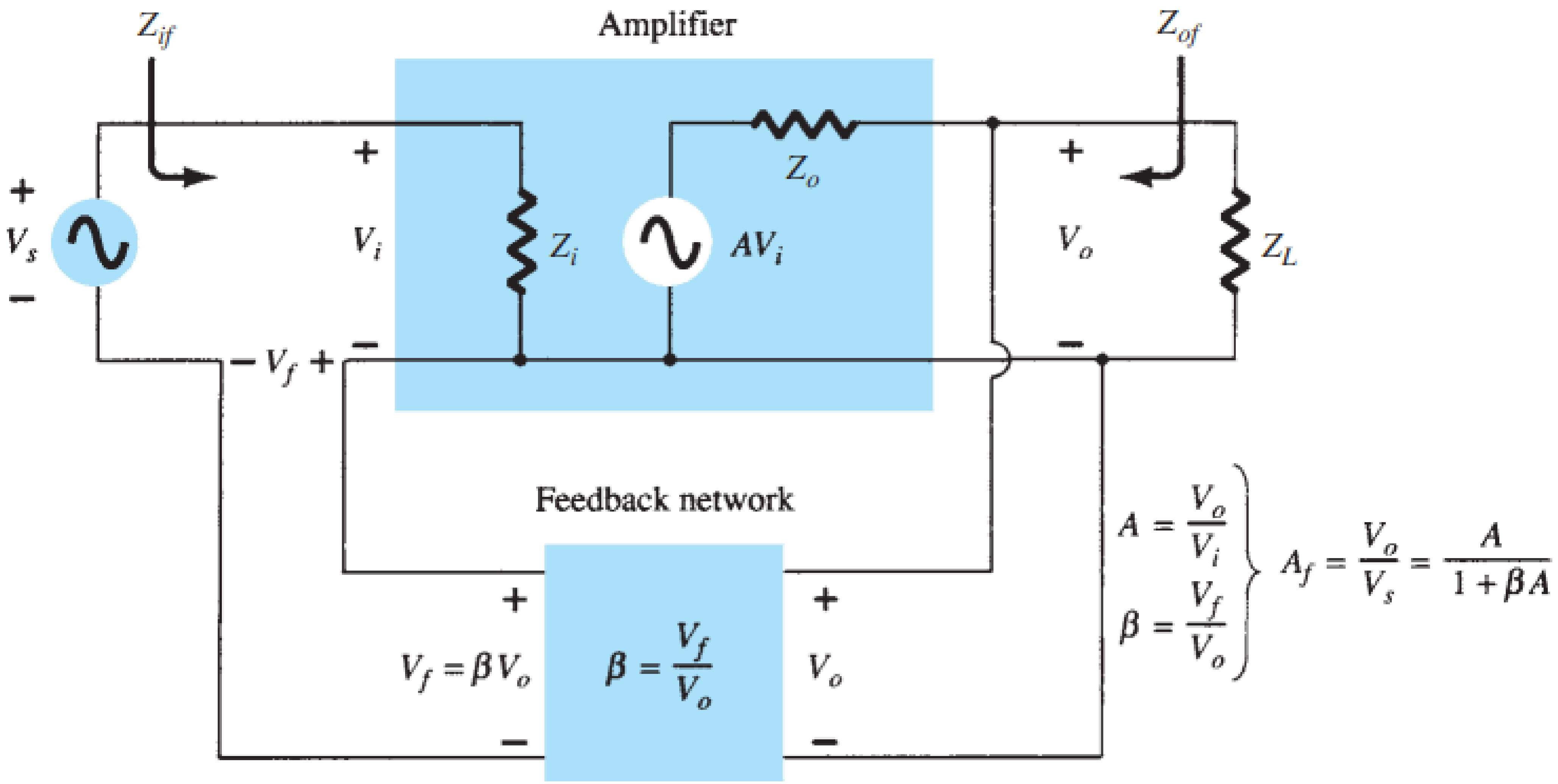
Problem 1: For a feedback amplifier A=200; fL=80 Hz; fH=60KHz
Given that only 2% of the output is returned to the input in opposition, find the modified value of gain and cut off frequency.



2. Voltage-shunt feedback

Input Impedance with Feedback

Voltage series feedback amplifier



The input impedance can be determined as follows:

$$I_i = \frac{V_i}{Z_i} = \frac{V_s - V_f}{Z_i} = \frac{V_s - \beta V_o}{Z_i} = \frac{V_s - \beta A V_i}{Z_i}$$

$$I_i Z_i = V_s - \beta A V_i$$

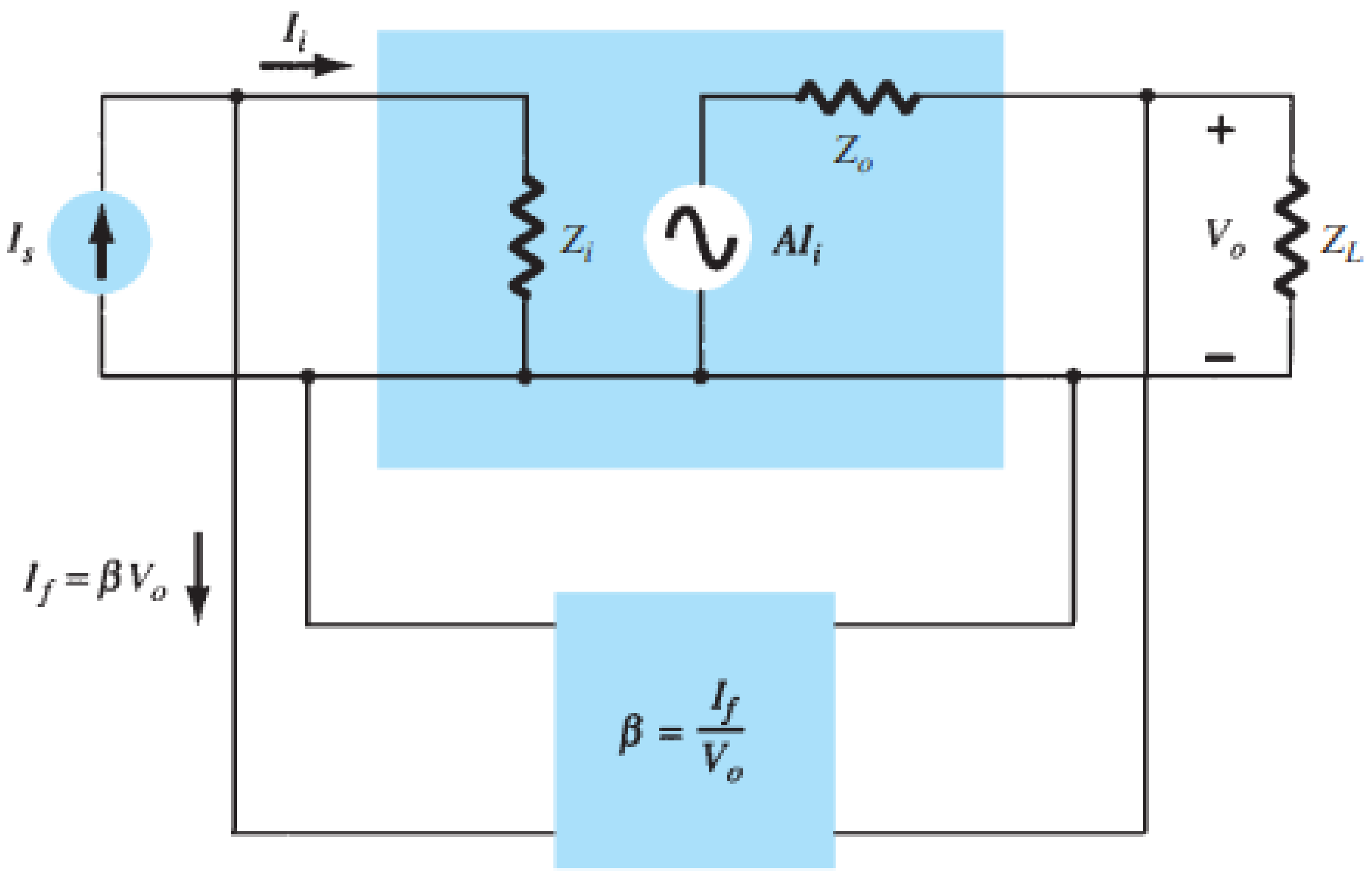
$$V_s = I_i Z_i + \beta A V_i = I_i Z_i + \beta A I_i Z_i$$

$$\boxed{Z_{if} = \frac{V_s}{I_i} = Z_i + (\beta A) Z_i = Z_i(1 + \beta A)}$$

The input impedance with series feedback is seen to be the value of the input impedance without feedback multiplied by the factor (1+ βA), and applies to both voltage-series and current-series configurations

Input Impedance with Feedback

Voltage shunt feedback amplifier



The input impedance can be determined to be

$$Z_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i + I_f} = \frac{V_i}{I_i + \beta V_o}$$

$$= \frac{V_i/I_i}{I_i/I_i + \beta V_o/I_i}$$

$$\boxed{Z_{if} = \frac{Z_i}{1 + \beta A}}$$

Output Impedance with Feedback

The output impedance for the connections of is dependent on whether voltage or current feedback is used. For voltage feedback, the output impedance is decreased, whereas current feedback increases the output impedance.

Voltage-Series Feedback

The output impedance is determined by applying a voltage V , resulting in a current I , with Vs shorted out (Vs = 0). The voltage V is then

$$V = IZ_o + AV_i$$

For $V_s = 0$, $\quad V_i = -V_f$

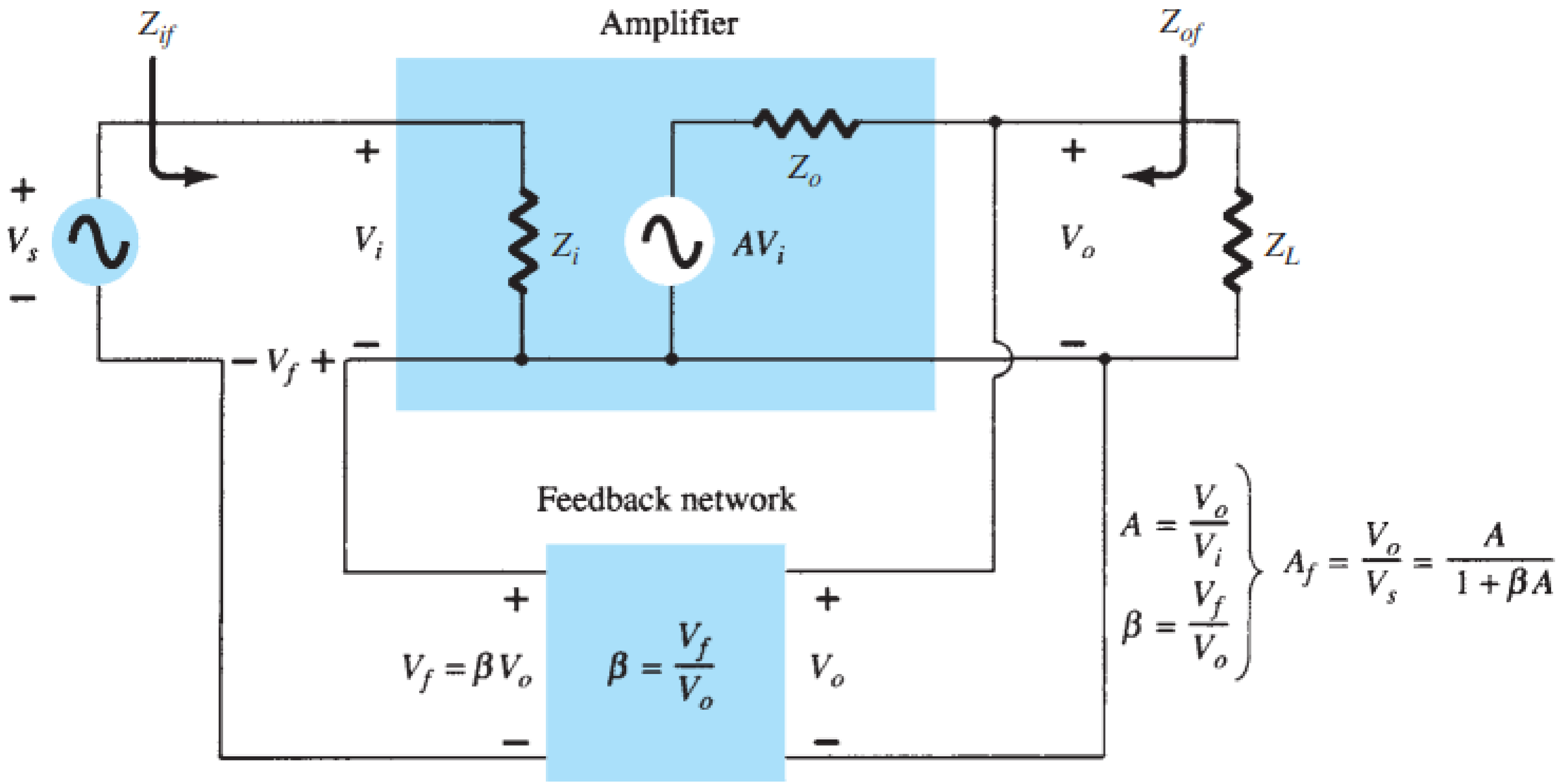
so that $\quad V = IZ_o - AV_f = IZ_o - A(\beta V)$

Rewriting the equation as

$$V + \beta AV = IZ_o$$

allows solving for the output impedance with feedback:

$$\boxed{Z_{of} = \frac{V}{I} = \frac{Z_o}{1 + \beta A}}$$

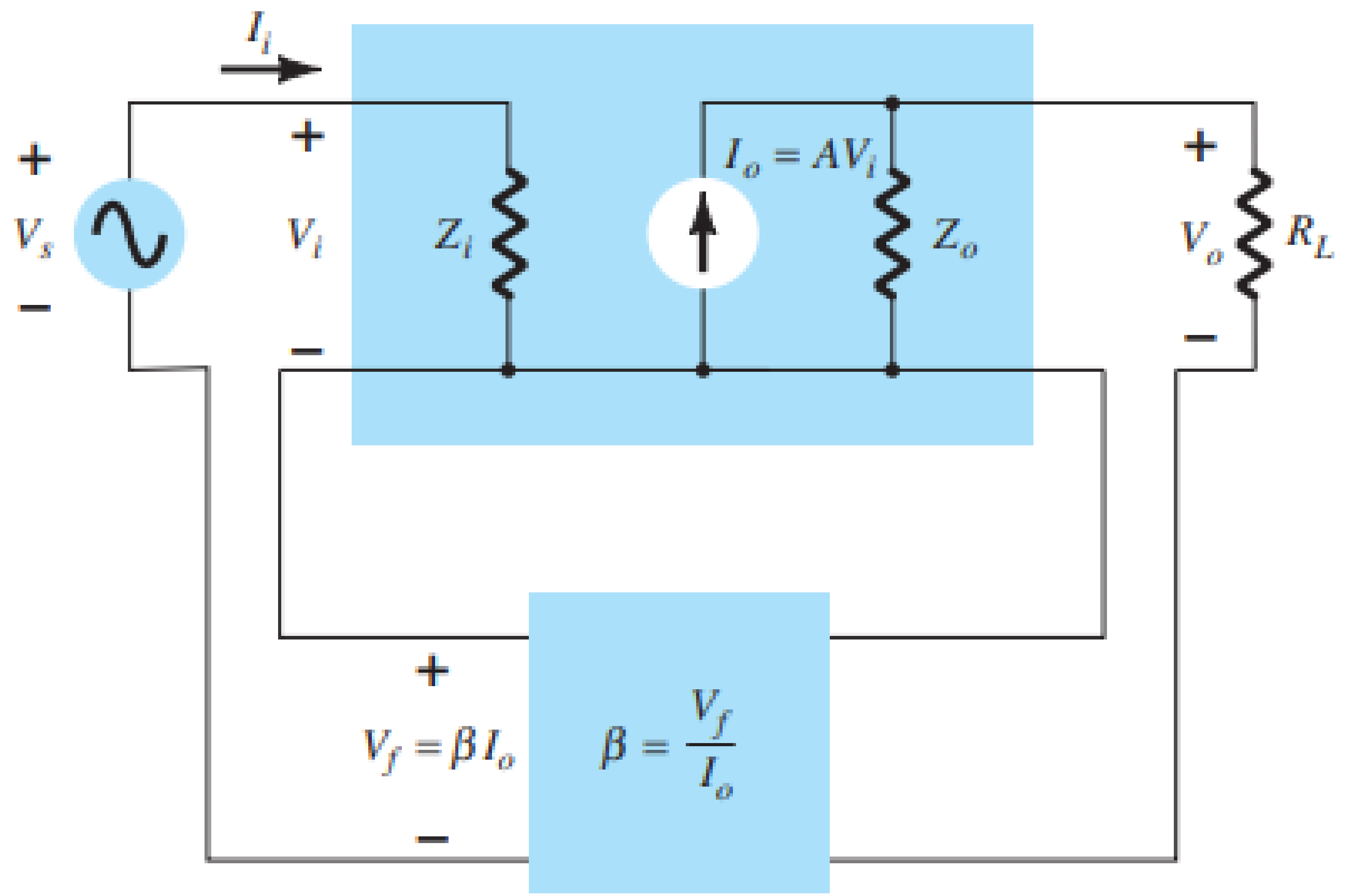


The expression is the same in case of Voltage shunt amplifier

Output Impedance with Feedback

Current Series Feedback

The output impedance with current-series feedback can be determined by applying a signal V to the output with V s shorted out, resulting in a current I , the ratio of V to I being the output connection with current-series feedback. For the output part of a current-series feedback connection shown in figure the resulting output impedance is determined as follows. With V s = 0,



$$V_i = V_f$$

$$I = \frac{V}{Z_o} - AV_i = \frac{V}{Z_o} - AV_f = \frac{V}{Z_o} - A\beta I$$

$$Z_o(1 + \beta A)I = V$$

$$\boxed{Z_{of} = \frac{V}{I} = Z_o(1 + \beta A)}$$

BJT: Frequency Considerations

Effect of Feedback Connection on Input and Output Impedance

	Voltage-Series	Current-Series	Voltage-Shunt	Current-Shunt
Z_{if}	$Z_i(1 + \beta A)$ (increased)	$Z_i(1 + \beta A)$ (increased)	$\frac{Z_i}{1 + \beta A}$ (decreased)	$\frac{Z_i}{1 + \beta A}$ (decreased)
Z_{of}	$\frac{Z_o}{1 + \beta A}$ (decreased)	$Z_o(1 + \beta A)$ (increased)	$\frac{Z_o}{1 + \beta A}$ (decreased)	$Z_o(1 + \beta A)$ (increased)

Reduction in Frequency Distortion

For a negative-feedback amplifier having βA>>1, the gain with feedback is $A_f = 1/\beta$. It follows from this that if the feedback network is purely resistive, the gain with feedback is not dependent on frequency even though the basic amplifier gain is frequency dependent. Practically, the frequency distortion arising because of varying amplifier gain with frequency is considerably reduced in a negative-voltage feedback amplifier circuit.

Reduction in Noise and Nonlinear Distortion

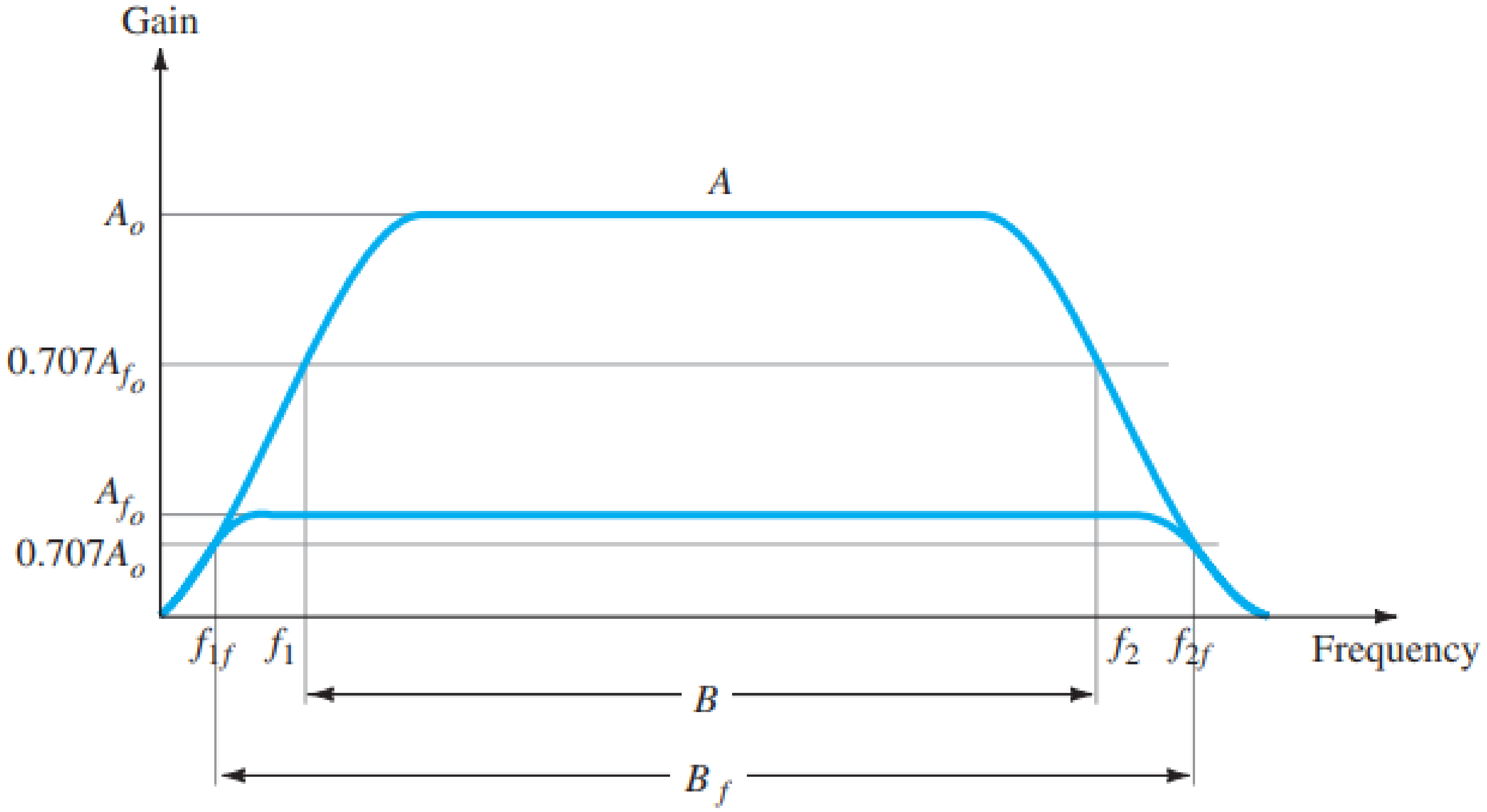
Signal feedback tends to hold down the amount of noise signal (such as power-supply hum) and nonlinear distortion. The factor (1+βA) reduces both input noise and resulting nonlinear distortion for considerable improvement. However, there is a reduction in over-all gain (the price required for the improvement in circuit performance). If additional stages are used to bring the overall gain up to the level without feedback, the extra stage(s) might introduce as much noise back into the system as that reduced by the feedback amplifier. This problem can be somewhat alleviated by readjusting the gain of the feedback amplifier circuit to obtain higher gain while also providing reduced noise sign

Effect of Negative Feedback on Gain and Bandwidth

$$A_f = \frac{A}{1 + \beta A} \cong \frac{A}{\beta A} = \frac{1}{\beta} \qquad \text{for } \beta A \gg 1$$

As long as βA>>1, the overall gain is approximately $1/\beta$. For a practical amplifier (for single low- and high-frequency breakpoints) the open-loop gain drops off at high frequencies due to the active device and circuit capacitances. Gain may also drop off at low frequencies for capacitively coupled amplifier stages. Once the open-loop gain A drops low enough and the factor βA is no longer much larger than 1.

Figure shows that the amplifier with negative feedback has more bandwidth ($A = B_f$) than the amplifier without feedback (3-dB). The feedback amplifier has a higher upper 3-dB frequency and smaller lower 3-dB frequency.



Gain Stability with Feedback

In addition to the b factor setting a precise gain value, we are also interested in how stable the feedback amplifier is compared to an amplifier without feedback.

$$\boxed{A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A}}$$

Differentiating

$$\left|\frac{dA_f}{A_f}\right| = \frac{1}{|1 + \beta A|}\left|\frac{dA}{A}\right|$$

$$\left|\frac{dA_f}{A_f}\right| \cong \left|\frac{1}{\beta A}\right|\left|\frac{dA}{A}\right| \qquad \text{for } \beta A \gg 1$$

This shows that magnitude of the relative change in gain $\left|\frac{dA_f}{A_f}\right|$ is reduced by the factor $|\beta A|$ compared to that without feedback $\left(\left|\frac{dA}{A}\right|\right)$.

Question : For a feedback amplified A=1000, $\frac{dA_f}{A_f}$ =0.01% ;dA= 10; Find the Transfer function of the feedback network.

FEEDBACK AMPLIFIER—PHASE AND FREQUENCY CONSIDERATIONS

- Negative feedback-The operation of a feedback amplifier in which the feedback signal was opposite to the input signal. In any practical circuit this condition occurs only for some mid-frequency range of operation.

- For an amplifier the gain will change with frequency, dropping off at high frequencies from the mid-frequency value.

- In addition, the phase shift of an amplifier will also change with frequency.

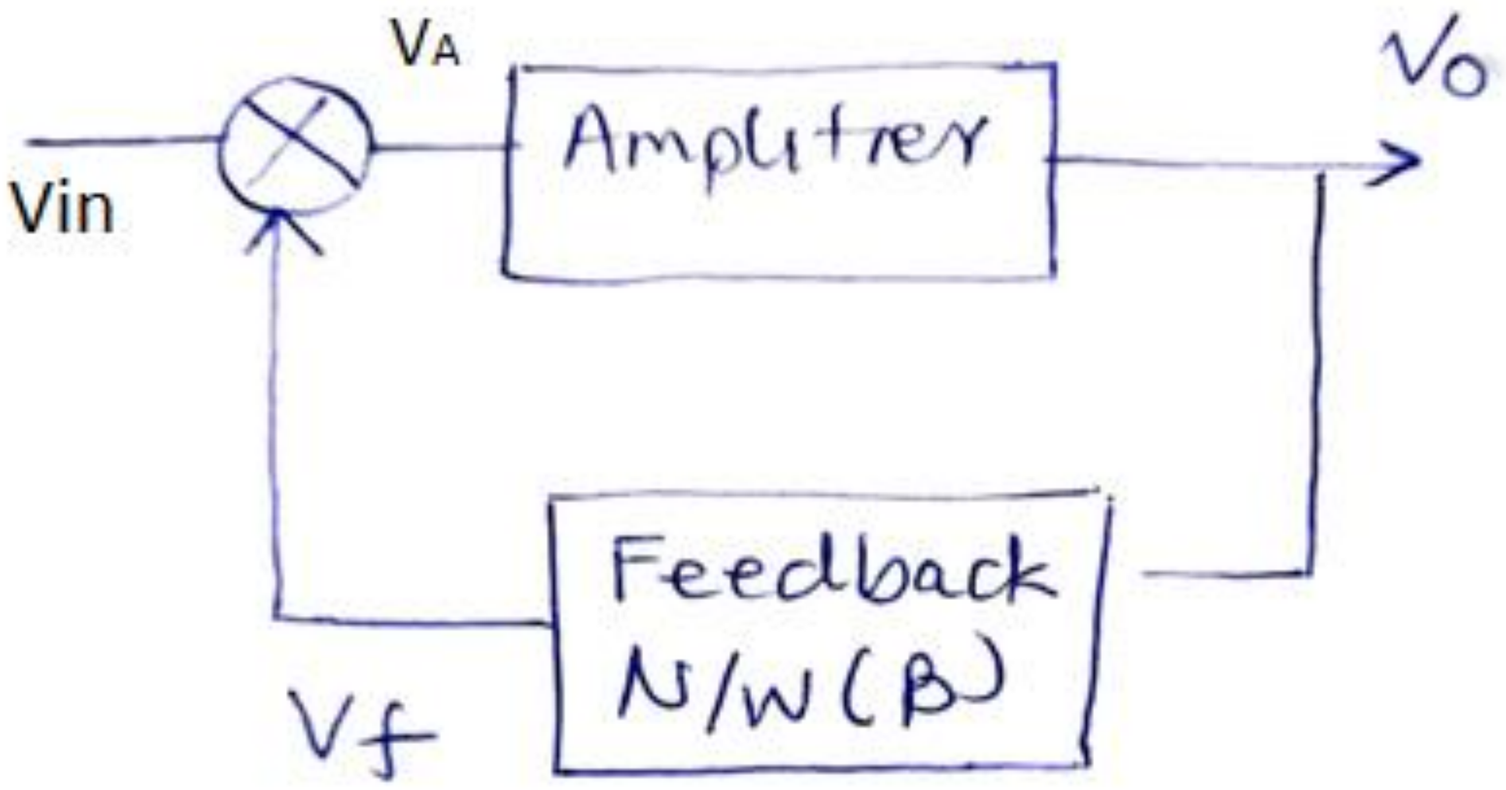
- If, as the frequency increases, the phase shift changes, then some of the feedback signal will add to the input signal. It is then possible for the amplifier to break into oscillations due to positive feedback.

- If the amplifier oscillates at some low or high frequency, it is no longer useful as an amplifier.

- Proper feedback-amplifier design requires that the circuit be stable at all frequencies, not merely those in the range of interest. Otherwise, a transient disturbance could cause a seemingly stable amplifier to suddenly start oscillating.

Oscillator

Converts DC to AC
Feedback amplifier (Positive feedback)



Apply Vin
$V_o = AV_A$
$V_f = \beta V_o$
$V_A = V_{in} + V_f = V_{in} + \beta V_o$
$V_o = A(V_{in} + \beta V_o)$
So, for Gain
$V_o(1 - A\beta) = A V_{in}$
$V_o/V_{in} = A/(1-AB)$

If $V_{in} = 0$
$V_A = V_f$
$V_o = AV_f$
$V_o = A\beta V_o$
$A\beta = 1$

- CE Configuration
- Phase shift 180 degrees
- Finite voltage gain

- Frequency selective element
- Allow fixed frequency to pass through
- Transfer function of feedback network

Oscillator

Barkhausen Criterion

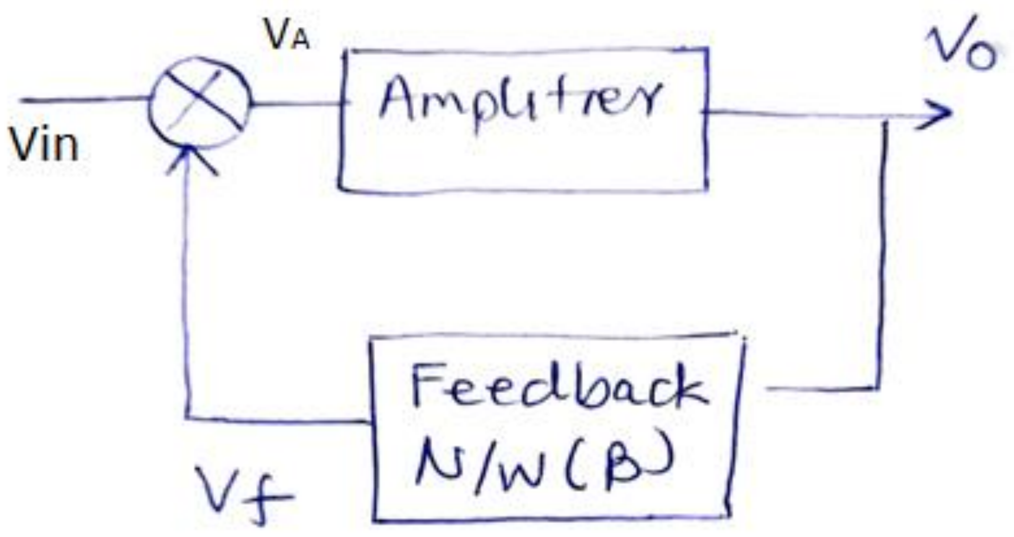
- A B=1
- Phase angle =0
- Sustained Oscillations

Oscillator: Types

(i) Harmonic oscillator

(ii)Relaxation Oscillator

Generally for a practical oscillator AB>1, so as to account for the nonlinear behaviour of the circuit



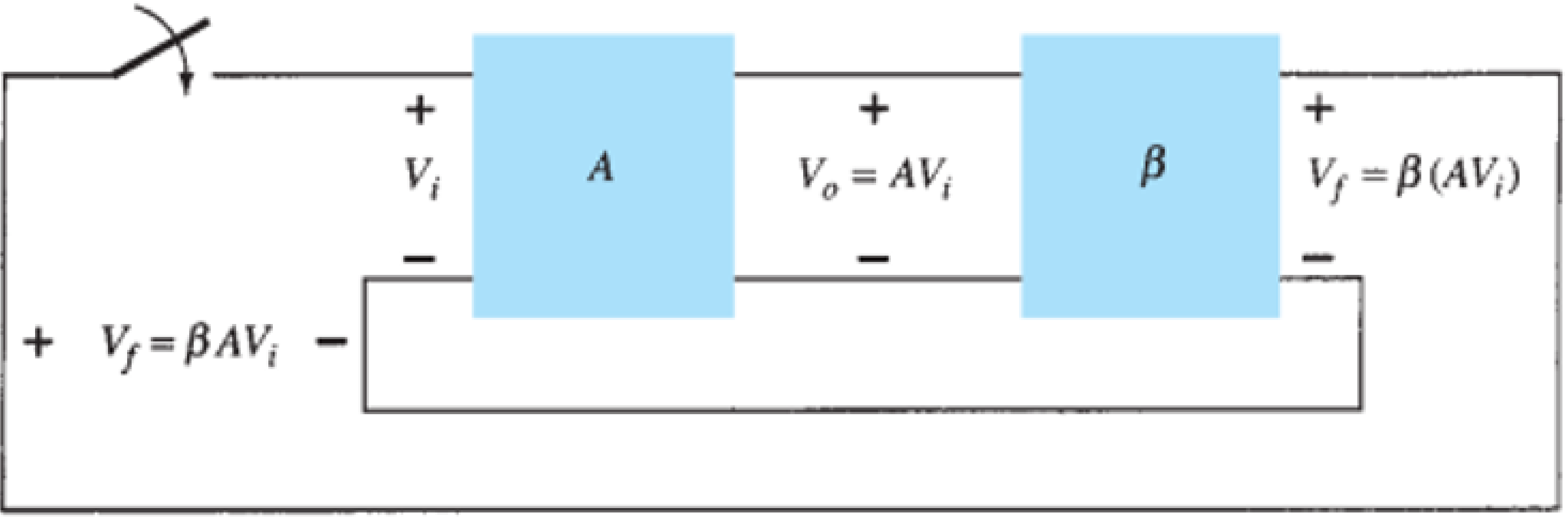
Frequency selective elements

RC, LC, Crystal

Oscillators

- The use of positive feedback that results in a feedback amplifier having closed-loop gain $|A_f|$ greater than 1 and satisfies the phase conditions will result in operation as an oscillator circuit. An oscillator circuit then provides a varying output signal.

- If the output signal varies sinusoidally, the circuit is referred to as a sinusoidal oscillator. If the output voltage rises quickly to one voltage level and later drops quickly to another voltage level, the circuit is generally referred to as a pulse or square-wave oscillator

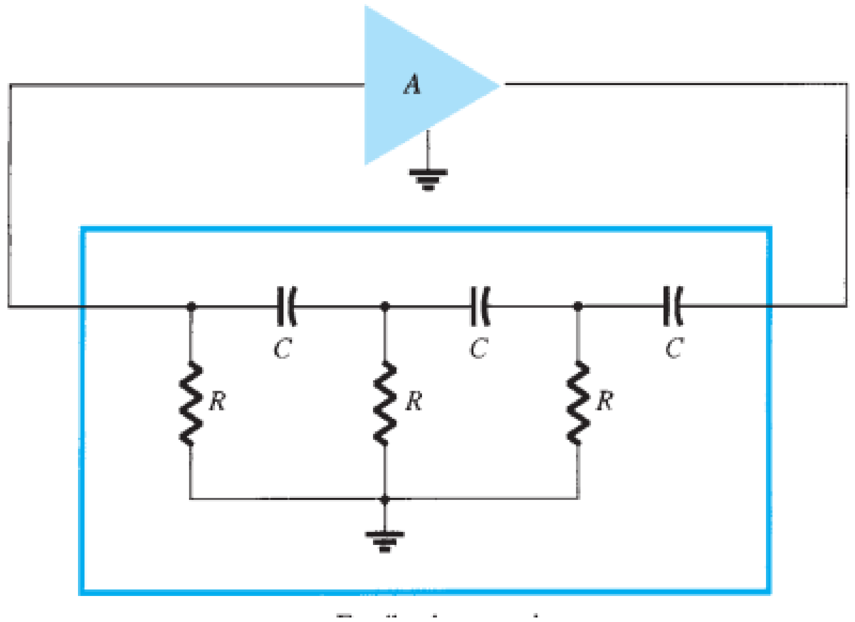


Let's consider that the switch at the amplifier input is open, no oscillation occurs. Consider that we have a fictitious voltage at the amplifier input Vi . This results in an output voltage Vo = AVi after the amplifier stage and in a voltage Vf = β(AVi) after the feedback stage. Thus, we have a feedback voltage Vf = $A = \beta$AVi, where βA is referred to as the loop gain. If the circuits of the base amplifier and feedback network provide βA of a correct magnitude and phase, Vfcan be made equal to V i

- In reality, no input signal is needed to start the oscillator going. Only the condition βA = 1 must be satisfied for self-sustained oscillations to result.
- In practice, βA is made greater than 1 and the system is started oscillating by amplifying noise voltage, which is always present.

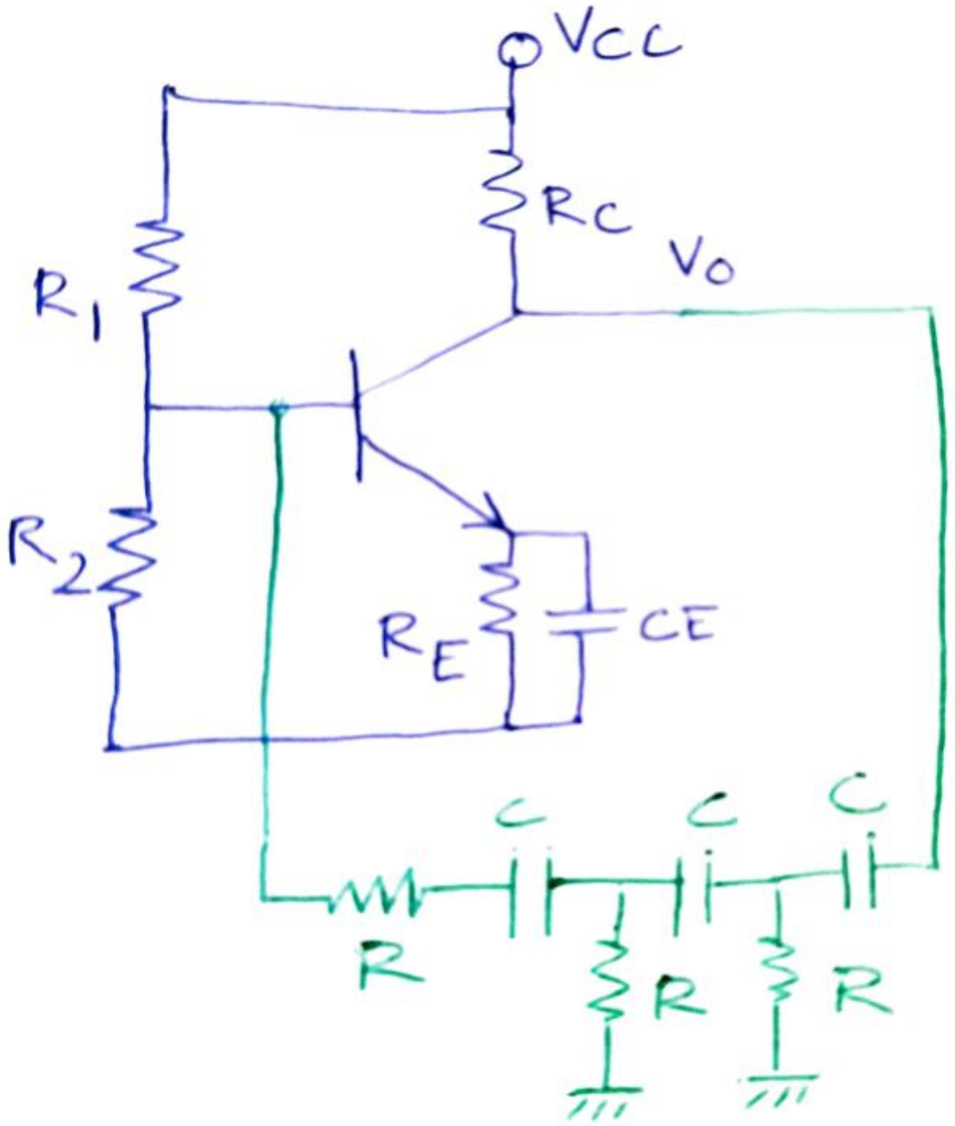
Then, when the switch is closed and the fictitious voltage Vi is removed, the circuit will continue operating since the feedback voltage is sufficient to drive the amplifier and feedback circuits, resulting in a proper input voltage to sustain the loop operation. The output waveform will still exist after the switch is closed if the condition βA = 1 is met. This is known as the Barkhausen criterion for oscillation.

Phase shift Oscillator

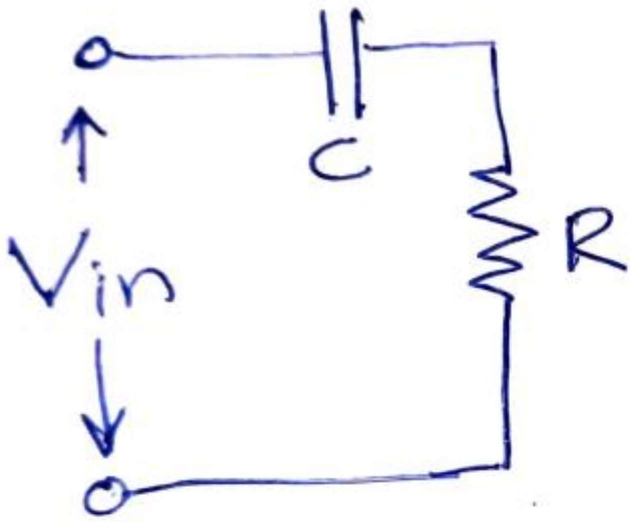


$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$\beta = \frac{1}{29}$$



Basic Circuit Diagram



$$V_o = \frac{R}{R+X_c} V_{in}$$

$$\frac{V_o}{V_{in}} = \frac{R}{R+\frac{1}{j\omega C}}$$

$$\phi = -\tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Each RC section shall provide a phase lag of 60 degrees
Phase lag introduced by amplifier is 180 degrees

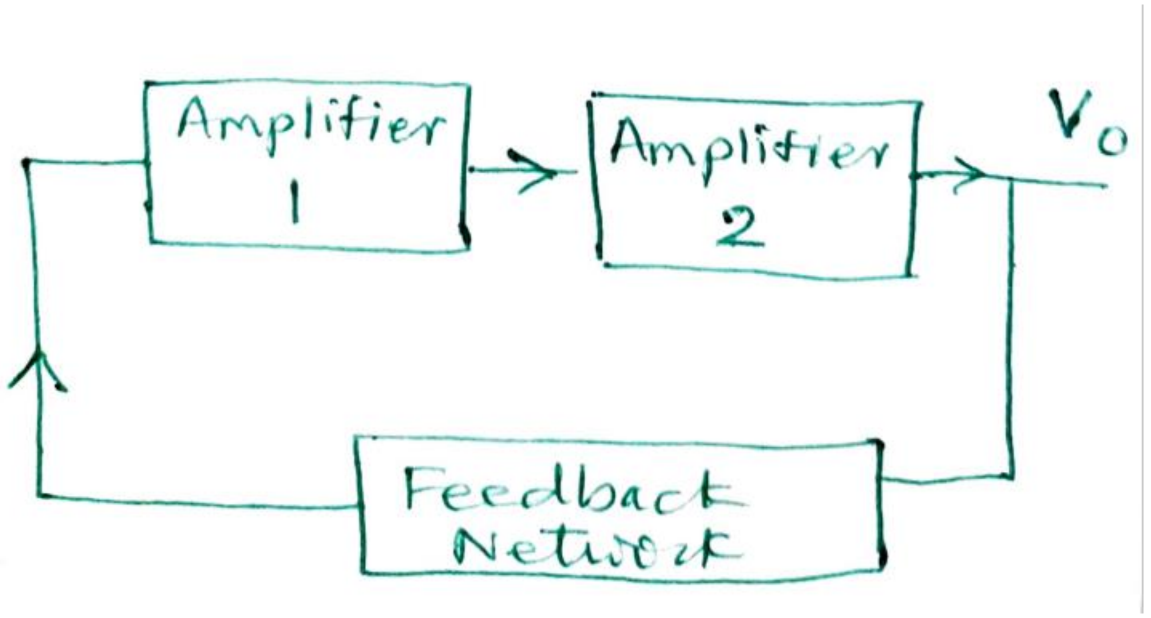
ADVANTAGES

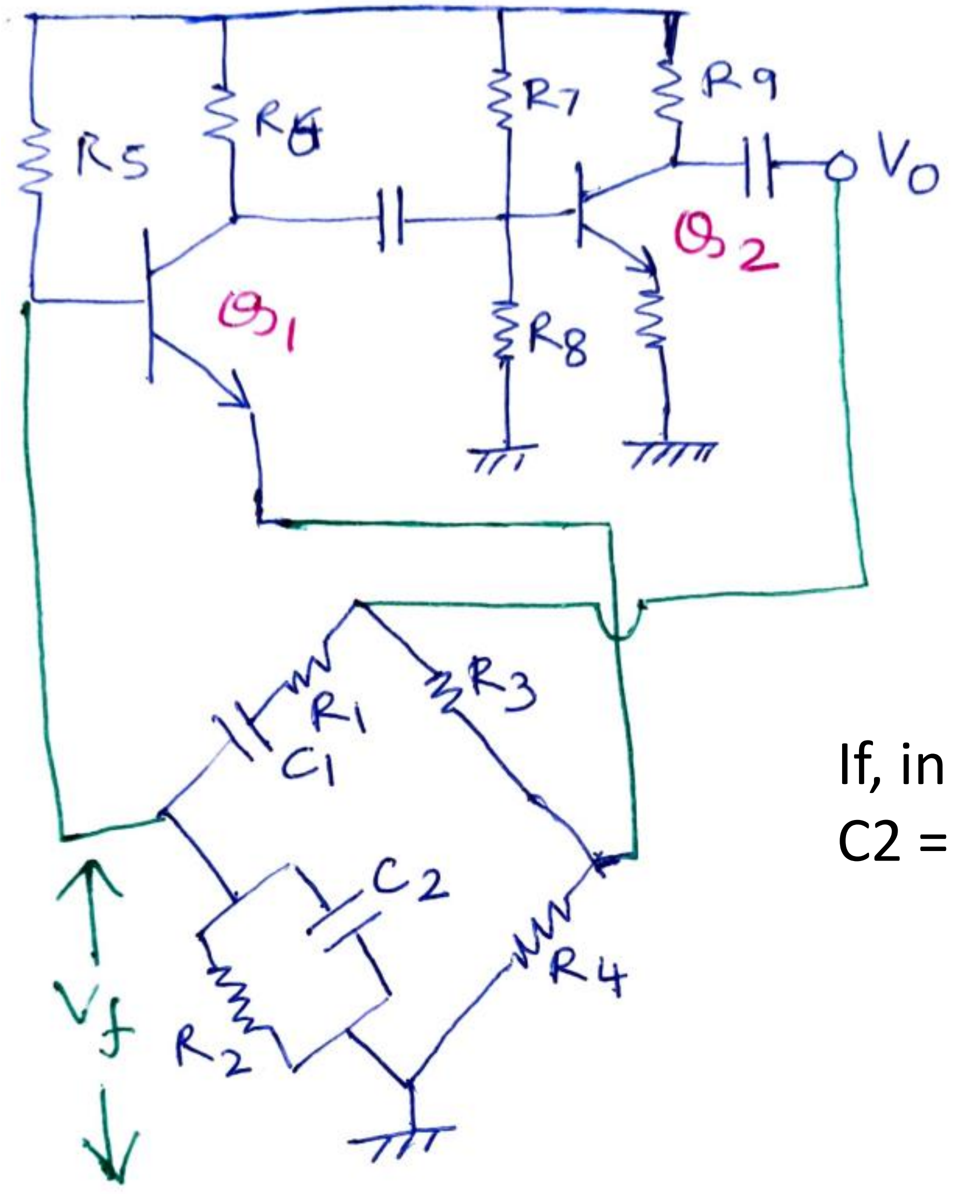
- Small size
- Efficiently used in audio range (20 Hz to 20KHz)
- Well suited for frequency below 10KHz
- Fixed frequency oscillator

DISADVANTAGES

- Low output
- Frequency stability is poor due to temperature and aging
- Require high gain, practically not possible

Wein Bridge Oscillator





$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

$$\boxed{f_o = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}}}$$

If, in particular, the values are R1 = R2 = R and C1 = C2 = C, the resulting Oscillator frequency is

$$\boxed{f_o = \frac{1}{2\pi RC}}$$

$$\boxed{\frac{R_3}{R_4} = 2}$$

Sailient features

- Audio frequency amplifier
- Two amplifiers are used
- Feedback network does not produce any phase lag
- Lead lag network is used as frequency network
- At fo no phase lag introduced by the feedback network
- At f> fo, phase lag is introduced
- At f<fo phase lead is introduced
- R and C elements are used in feedback network
- Overall gain is very high
- Provides variable frequency range
- Complex circuit, need to balance the bridge

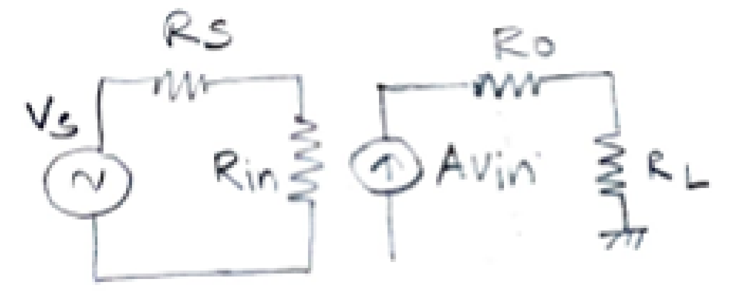
Thus a ratio of R3 to R4 greater than 2 will provide sufficient loop gain for the circuit to oscillate at the calculated frequency

Class of Amplifiers

Amplifier classes represent the amount the output signal varies over one cycle of operation for a full cycle of input signal.

$$Power = \frac{V^2}{R_L} = I^2 R_L$$

- In small-signal amplifiers, the main factors are usually amplification linearity and magnitude of gain. Since signal voltage and current are small in a small-signal amplifier, the amount of power-handling capacity and power efficiency are of little concern.
- A voltage amplifier provides voltage amplification primarily to increase the voltage of the input signal and current amplifier increases the current to the load.
- Large-signal or power amplifiers, on the other hand, primarily provide sufficient power to an output load to drive a speaker or other power device, typically a few watts to tens of watts.
- Here we discuss the amplifier circuits used to handle large-voltage signals at moderate to high current levels. The main features of a large-signal amplifier are the circuit's power efficiency, the maximum amount of power that the circuit is capable of handling, and the impedance matching to the output device.



$$V_{out} = \frac{R_L}{R_o + R_L} AV_{in}$$

Power =?

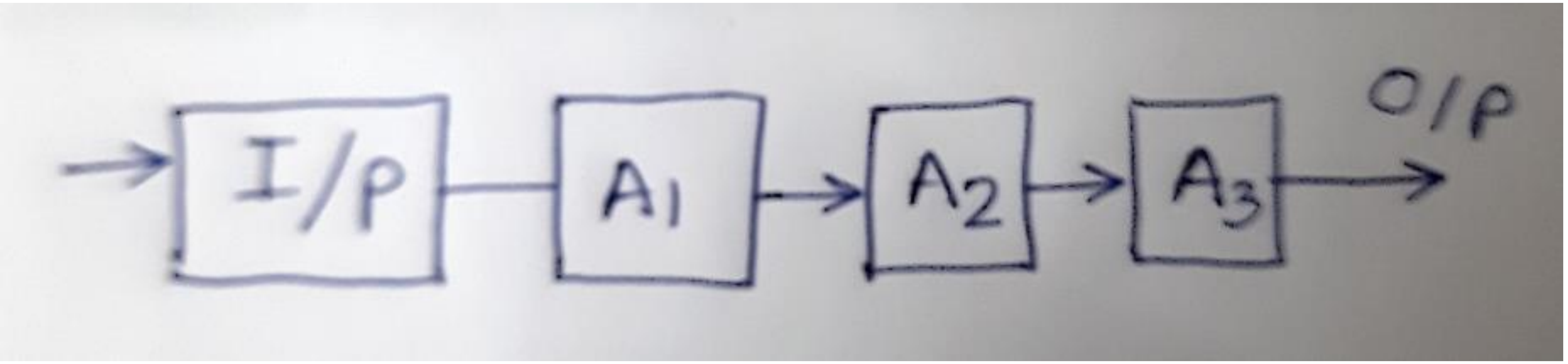
current source has high internal resistance

Both the Voltage amplifier as well as current amplifiers
- **work in linear region of the characteristics**
- **but, are not able to deliver high power**

$$I_L = \frac{R_o . AI_{in}}{R_o + R_L}$$

Power =?

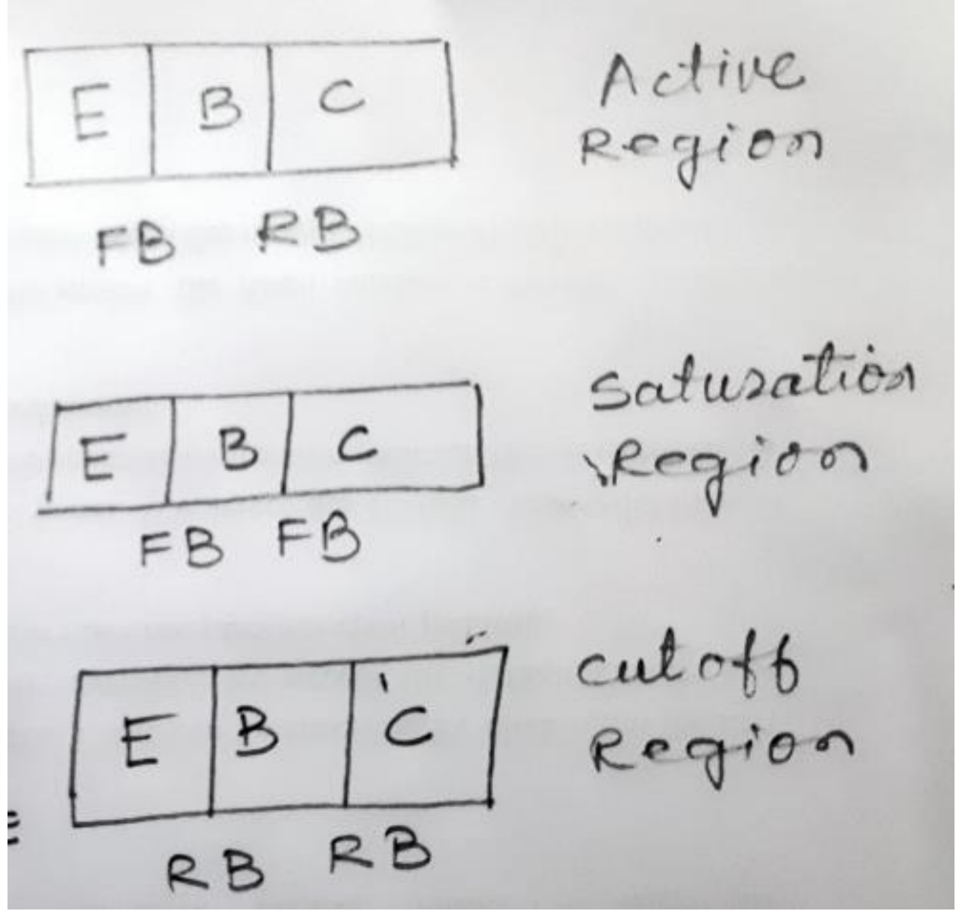
Difference between small signal amplifiers and Power amplifiers

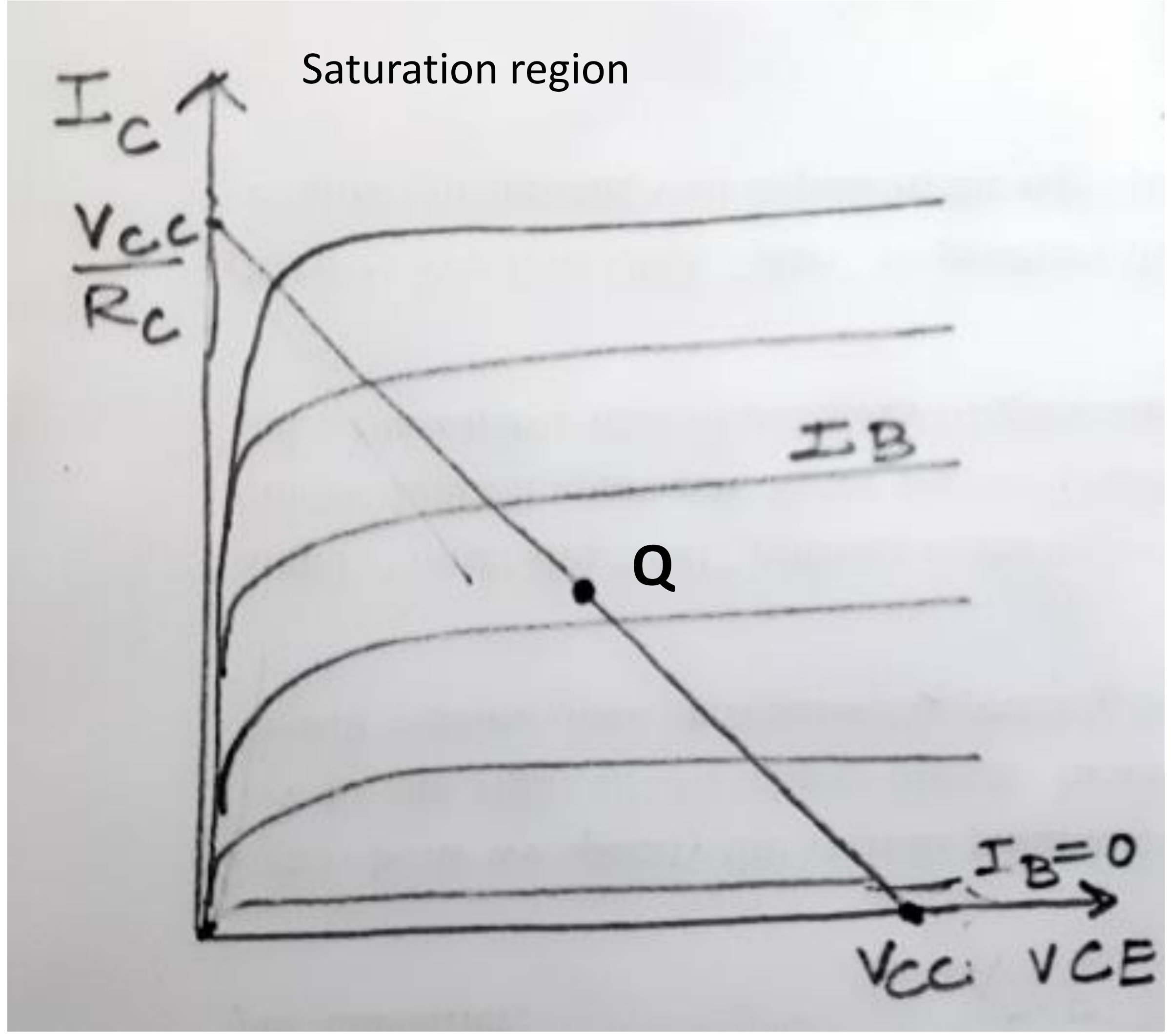


Power amplifiers handle large power signals.
They have inputs of the order of volts.

Small signal amplifiers	Power amplifiers
Signal strength is small	Signal strength is large
Active, Linear	Non-linear, Linear
Gain, input impedance, Output impedance	Distortion, Conversion efficiency
Analysis method: Mathematical, Graphical	Analysis method: Graphical

Types of Power Amplifiers





Cut off region

Operating Point /Quiescent point is f (I_C, V_{CE})

Classification

(i) Class A amplifiers

Linear region, middle of the DC load line

(ii) Class B amplifier

Non Linear region, Top/Bottom region of the load line

(iii) Class AB Amplifier

Slightly above saturation/ cut-off region

(i) Class C amplifier

Deep Cut-off region

Distortions

Harmonic Distortions

$$I_C = I_{CO} + I'_{CO} + I'_{C1}\sin\omega_1 t + I'_{C2}\sin\omega_2 t + \ldots$$

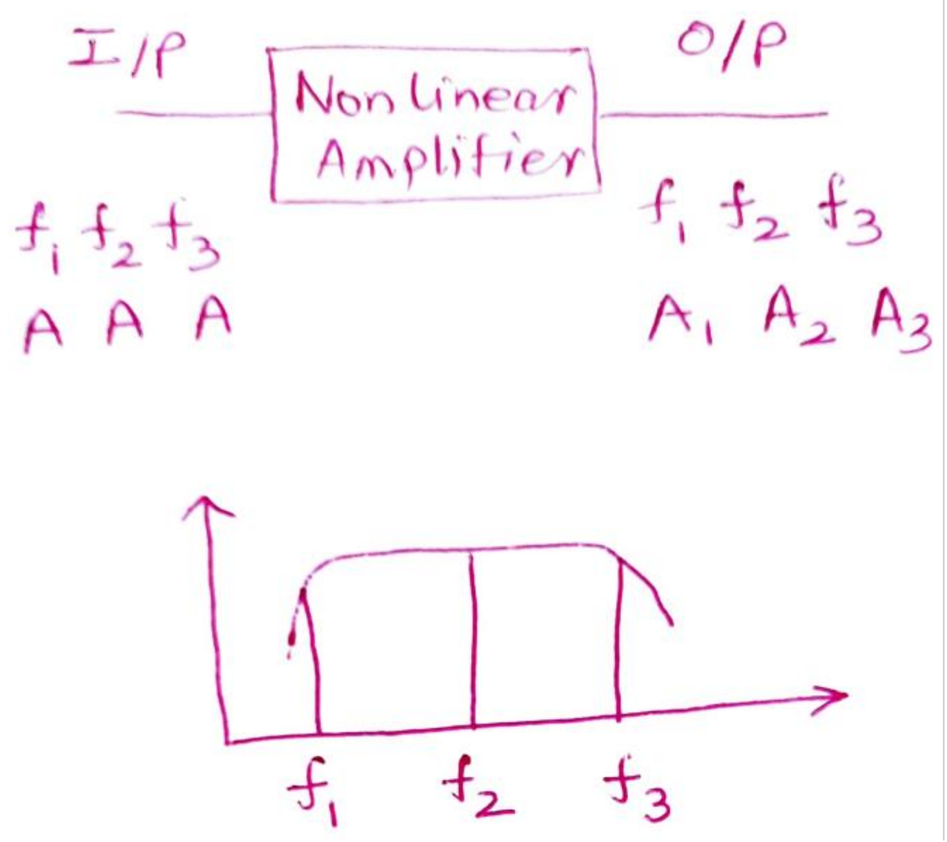
$$I_C = \underbrace{(I_{CO} + I'_{CO})}_{\text{DC}} + \underbrace{I'_{C1}\sin\omega_1 t}_{\text{fundamental freq.}} + \underbrace{I'_{C2}\sin\omega_2 t}_{\text{second harmonics}} + \ldots$$

$\omega_2 = 2\omega_1$

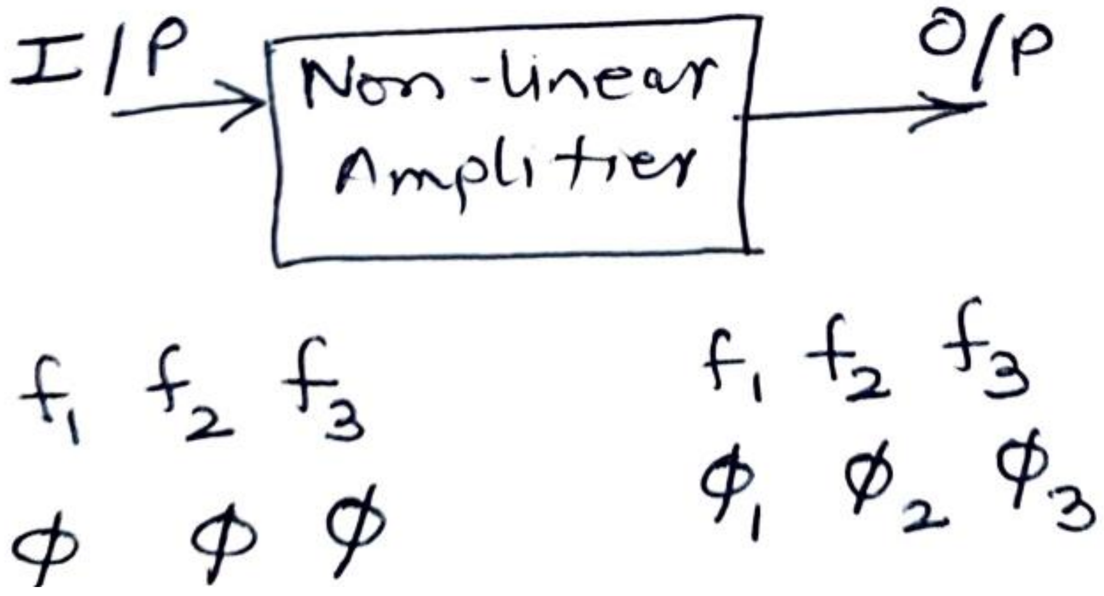
- If amplifier operates in the linear region, there is no change in the frequency of the output signal vis a vis the input signal. Also, if input is sinusoidal, output is also sinusoidal

- If the amplifier operates in non-linear operation, the output is a non-sinusoidal signal, which as per the Fourier series can be expressed in terms of fundamental frequency and it's harmonics $f_o, 2\,f_o, 3\,f_o$.......... (f_o= fundamental frequency)

(i) Frequency Distortions

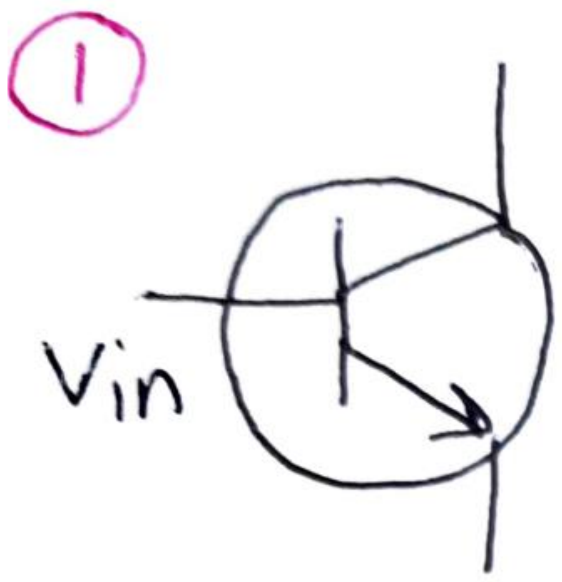


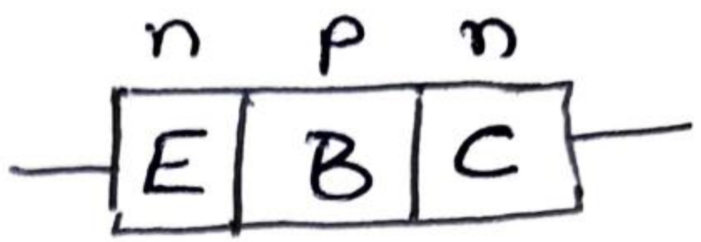
(ii) Phase distortions



Cross-over Distortions

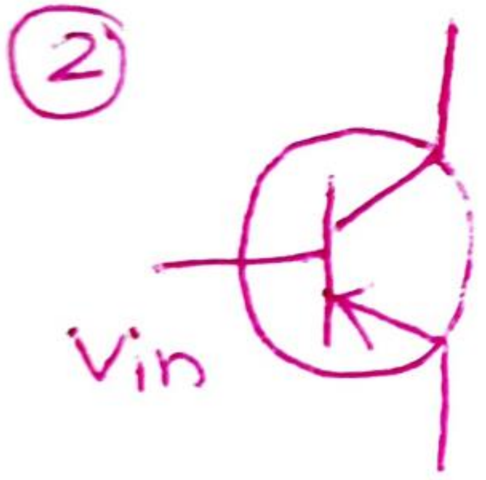
1





If $V_{in} > 0.7\,V$
then BE jun. → F.B.
else R.B.

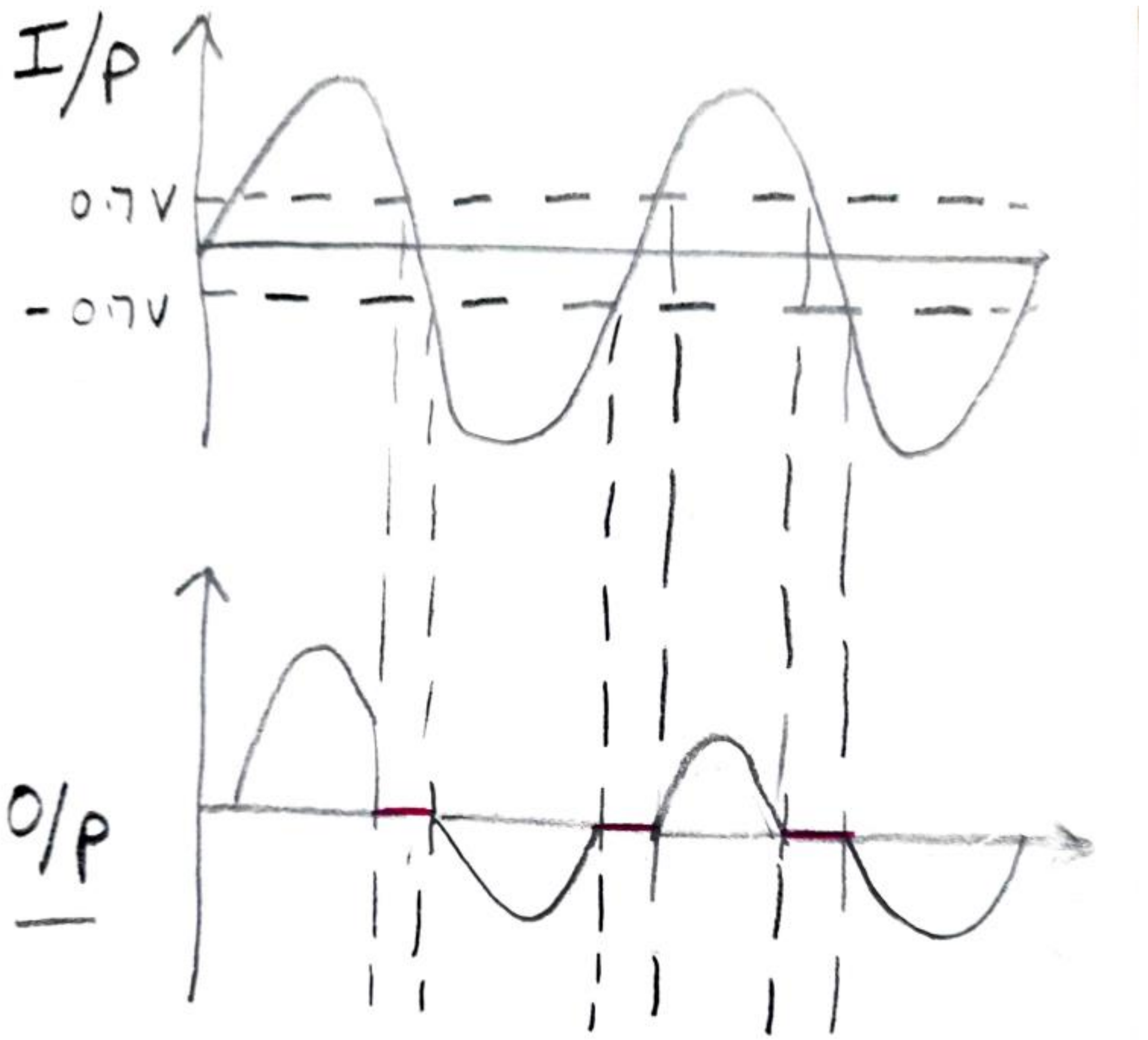
2



If $V_{in} < -0.7\,V$
them EB jun → F.B
else R.B.

Class B amplifiers

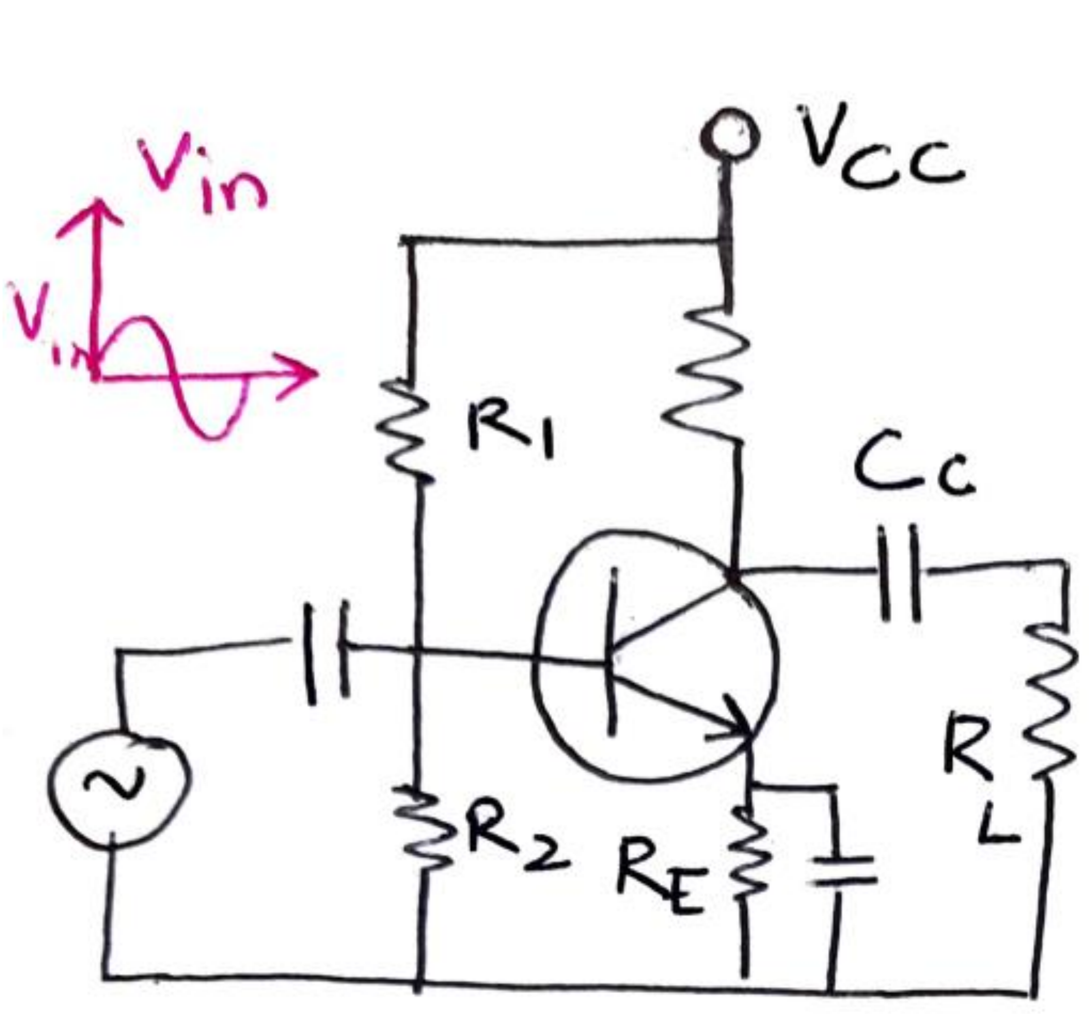
If $-0.7\,V < V_{in} < 0.7\,V$ then, E-B junction is Reverse biased



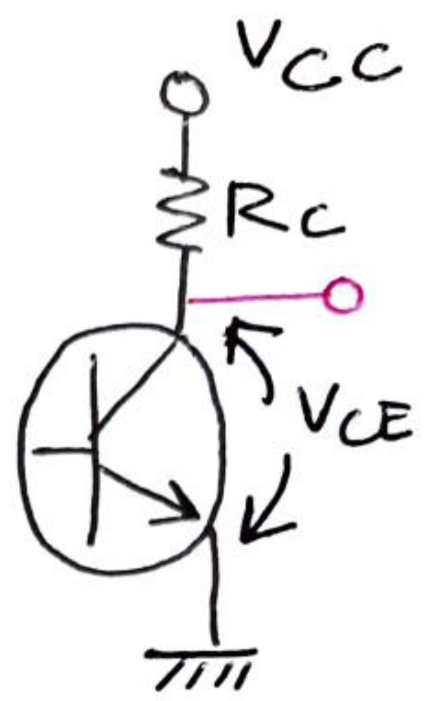
CROSS OVER DISTORTION

Dead zone: Range of input voltage within which output is zero

CLASS A Amplifiers

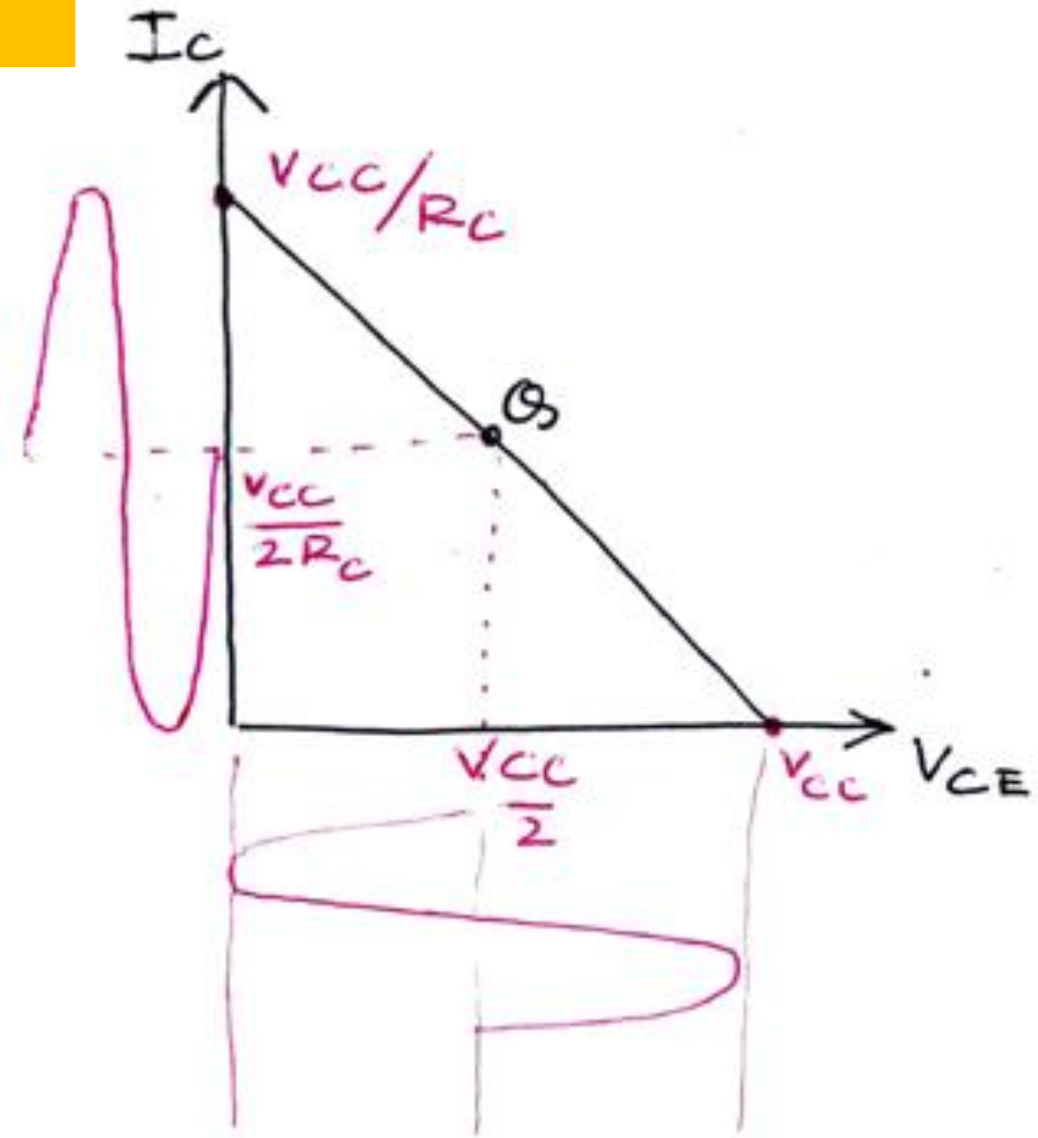


Circuit Diagram



$V_{CC} = I_C R_C + V_{CE}$

$V_{CE} = V_{CC} - I_C R_C$



Power Drain: For the amplifier to be operational DC biasing is necessary. Even if Vin =0, there shall be a power dissipation in the circuit which is given by P = I_{CQ} x V_{CQ}. This is the biggest disadvantage of the amplifier.

CASE 1

If Vin +ve cycle

I_B increases

$I_C = \beta I \Rightarrow I$ incr.

$V_{CE} = V_{CC} - I_C R_C$

Case 2

If Vin -ve cycle

I_B decreases

I_C decreases

$V_{CE} = V_{CC} - I_C R_C$

CLASS A Amplifiers

Characteristics

1.Output current flows for the full cycle

2.The output does not have linear distortions

3.Nature of output same as input

4.Q point lies in the middle of the load line

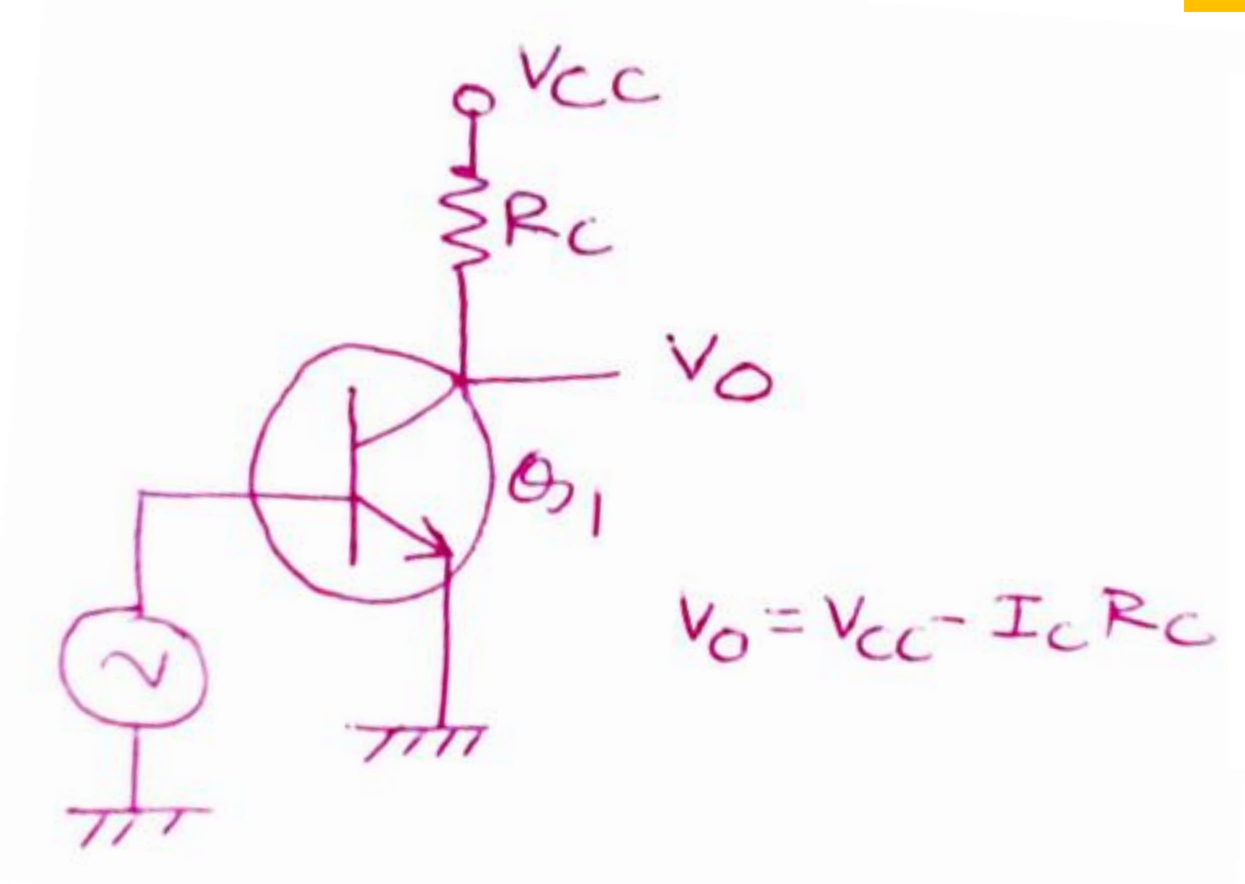
5.Power dissipation is high when the input is not applied.

6.The amplifier has low conversion efficiency

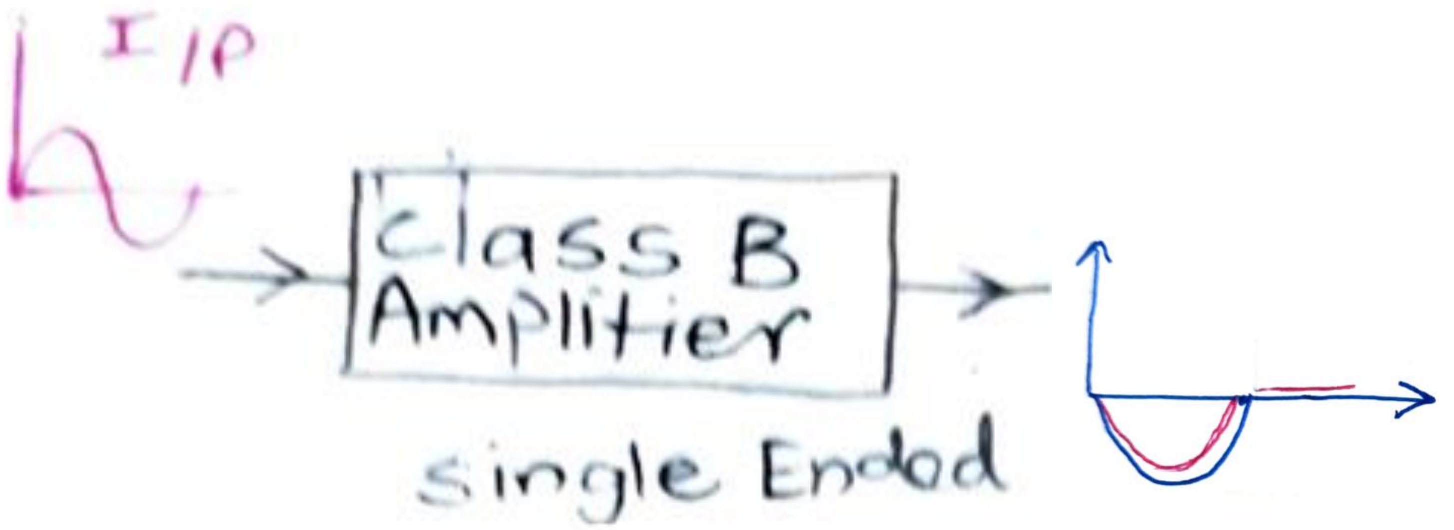
7.As the operation is in linear region the output is undistorted

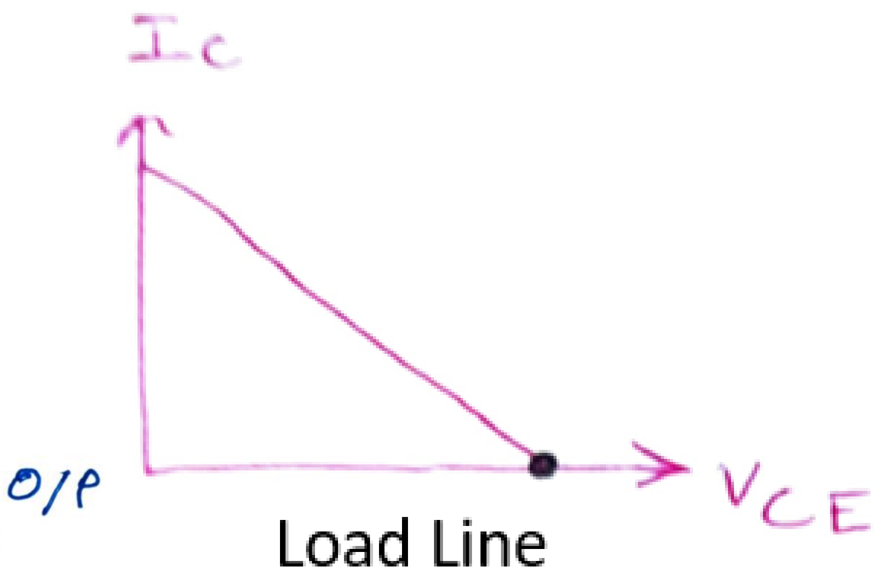
8.Conduction angle is 360 degrees

CLASS B Amplifiers



Single ended Class B Amplifier





CASE 1

Positive cycle

$Q_1 \rightarrow$ J_E FB, J_C RB — Active

Negative cycle

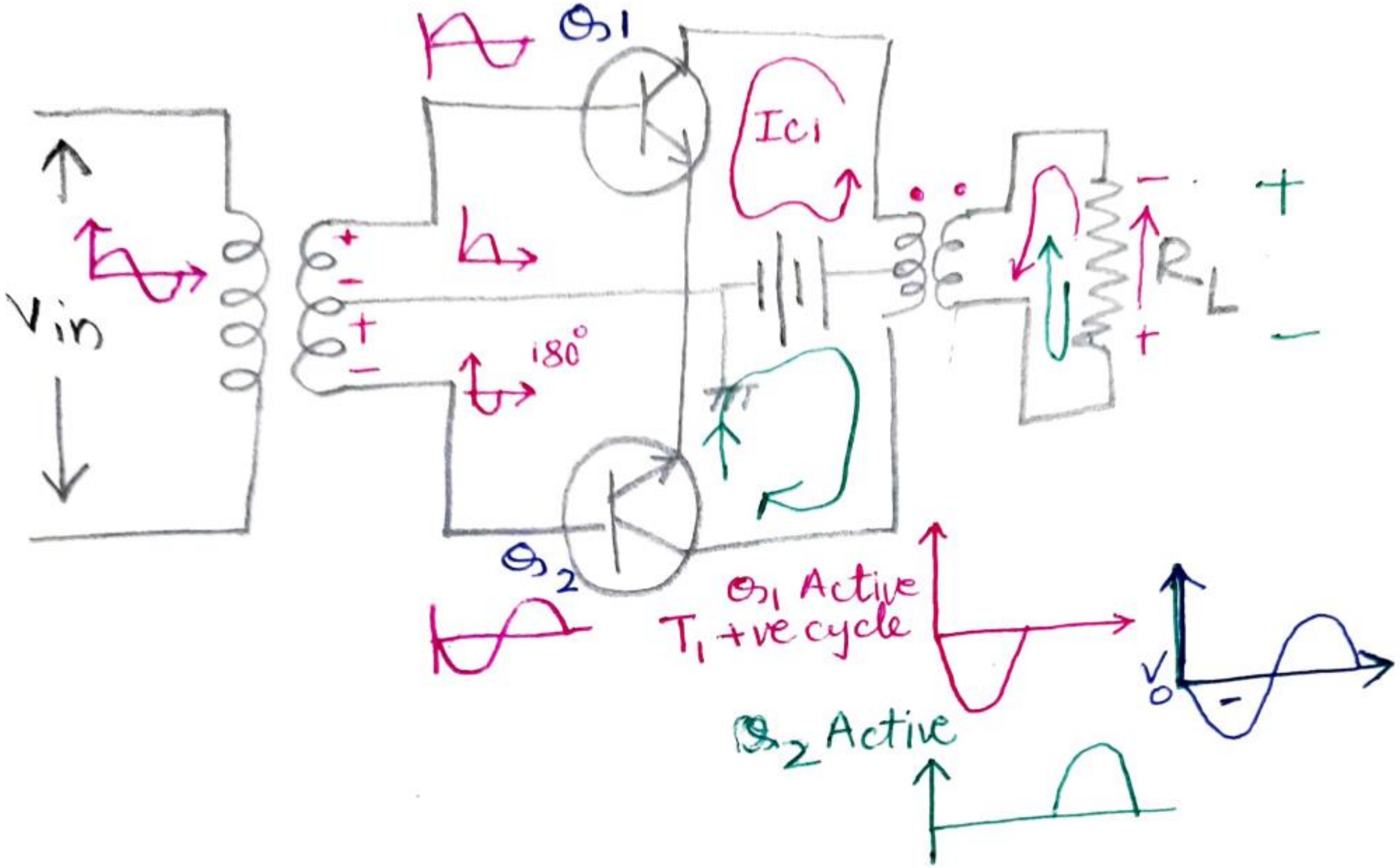
$Q_1 \rightarrow$ J_E RB, J_C RB

Salient features

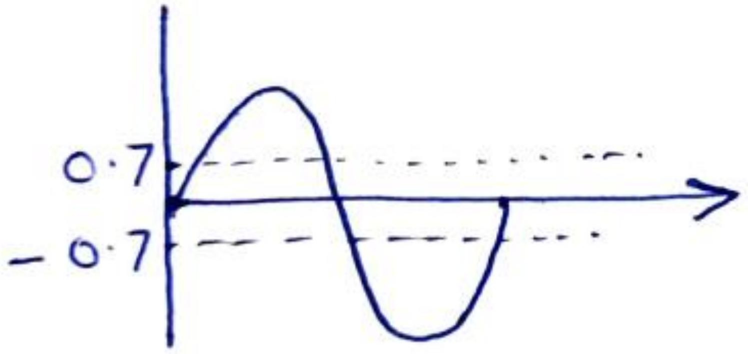
1. Q point is in Cut-off region.
2. No biasing is used.
3. Conversion efficiency is high.
4. Conduction angle is 180 degrees.
5. No power drain is encountered.
6. No power dissipation.
7. It has a lot of distortion.
8. Information of only one half of the cycle is obtained.

CLASS B Amplifiers

Class B Push pull amplifiers



- Two transistors are used: both are npn types
- Push pull mode is used
- Complete information reaches the output, as two BJTs used.
- Transformers (Centre tapped) are used
- Efficiency is high (78.5%)



Cross-over distortion

CLASS B Amplifiers

Advantages

1. Efficiency is high
2. No DC component at output
3. Even harmonics are cancelled
4. Power drain is absent

Disadvantages

1. Relatively large size and cost
2. Humming effect
3. Performance at low frequency is poor
4. Cross over distortion is present
5. If load is high transformer coupling cannot be used

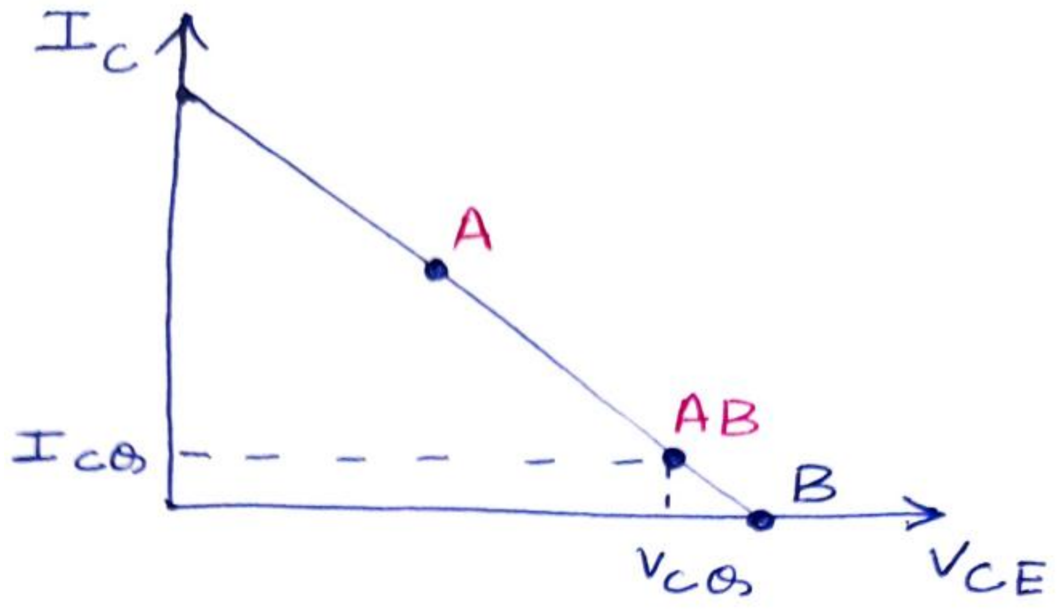
CLASS AB Amplifiers

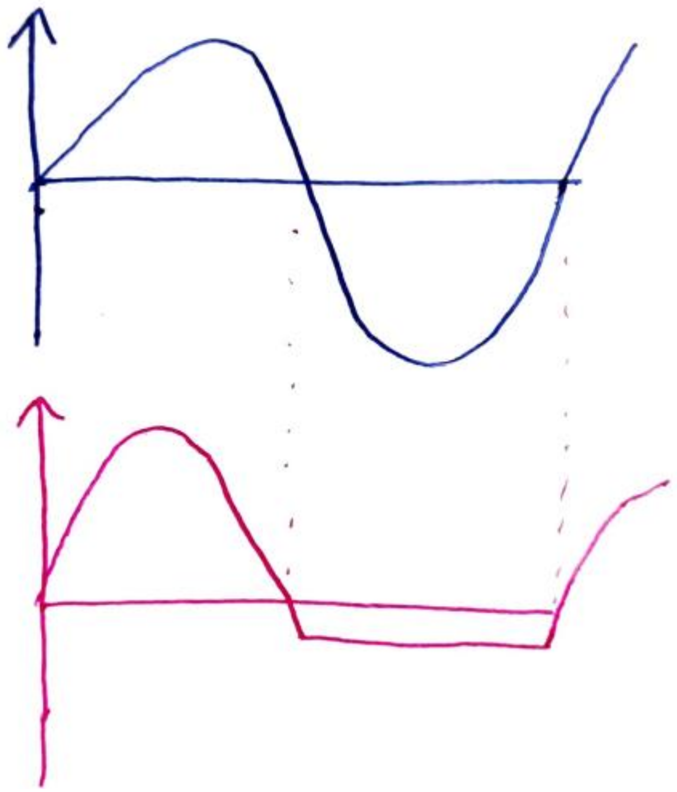
Class A - Conversion efficiency is Low: Power drain is high: Distortions are absent

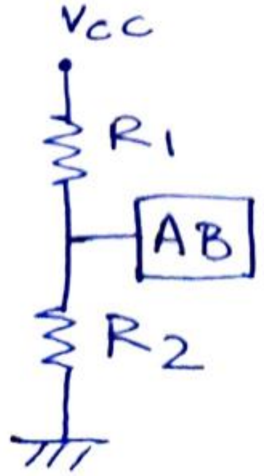
Class B - Conversion efficiency is high : Power drain is low: Distortions are present

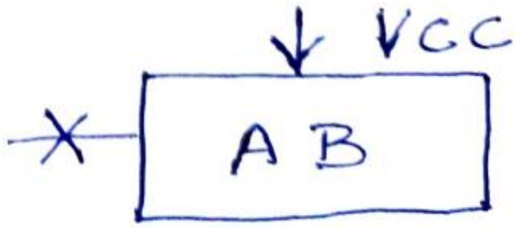
Salient features

- Q point lies slightly above the cut-off region
- Two BJTs are used (One npn and other pnp)
- Q point lies between Class A and Class B
- It conducts between 180 and 360 degrees
- Biasing voltage is applied such that the BJTs are slightly above the cut off region
- Biasing can be given with the help of resistor or the diodes









CLASS AB Amplifiers

Hartley Oscillator

Practical Feedback Circuits

Voltage series feedback amplifier

The emitter-follower circuit of Fig. provides voltage-series feedback. The signal voltage Vs is the input voltage Vi . The output voltage Vo is also the feedback voltage in series with the input voltage. The amplifier, as shown in Fig. , provides the operation with feedback. The operation of the circuit without feedback provides Vf = 0, so that

$$A = \frac{V_o}{V_s} = \frac{h_{fe} I_b R_E}{V_s} = \frac{h_{fe} R_E (V_s / h_{ie})}{V_s} = \frac{h_{fe} R_E}{h_{ie}}$$

and

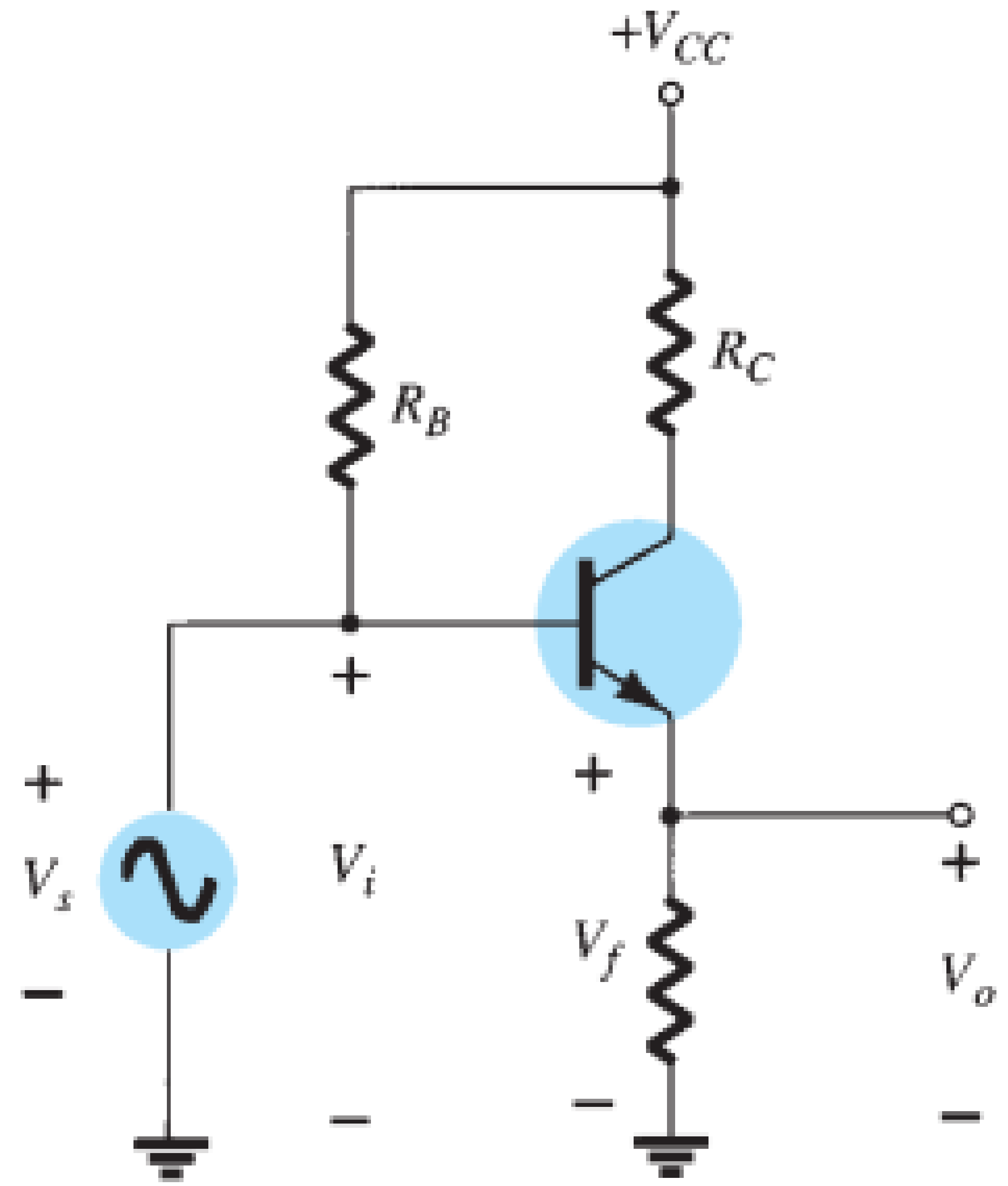
$$\beta = \frac{V_f}{V_o} = 1$$

The operation with feedback then provides that

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A} = \frac{h_{fe} R_E / h_{ie}}{1 + (1)(h_{fe} R_E / h_{ie})}$$
$$= \frac{h_{fe} R_E}{h_{ie} + h_{fe} R_E}$$

For $h_{fe} R_E \gg h_{ie}$,

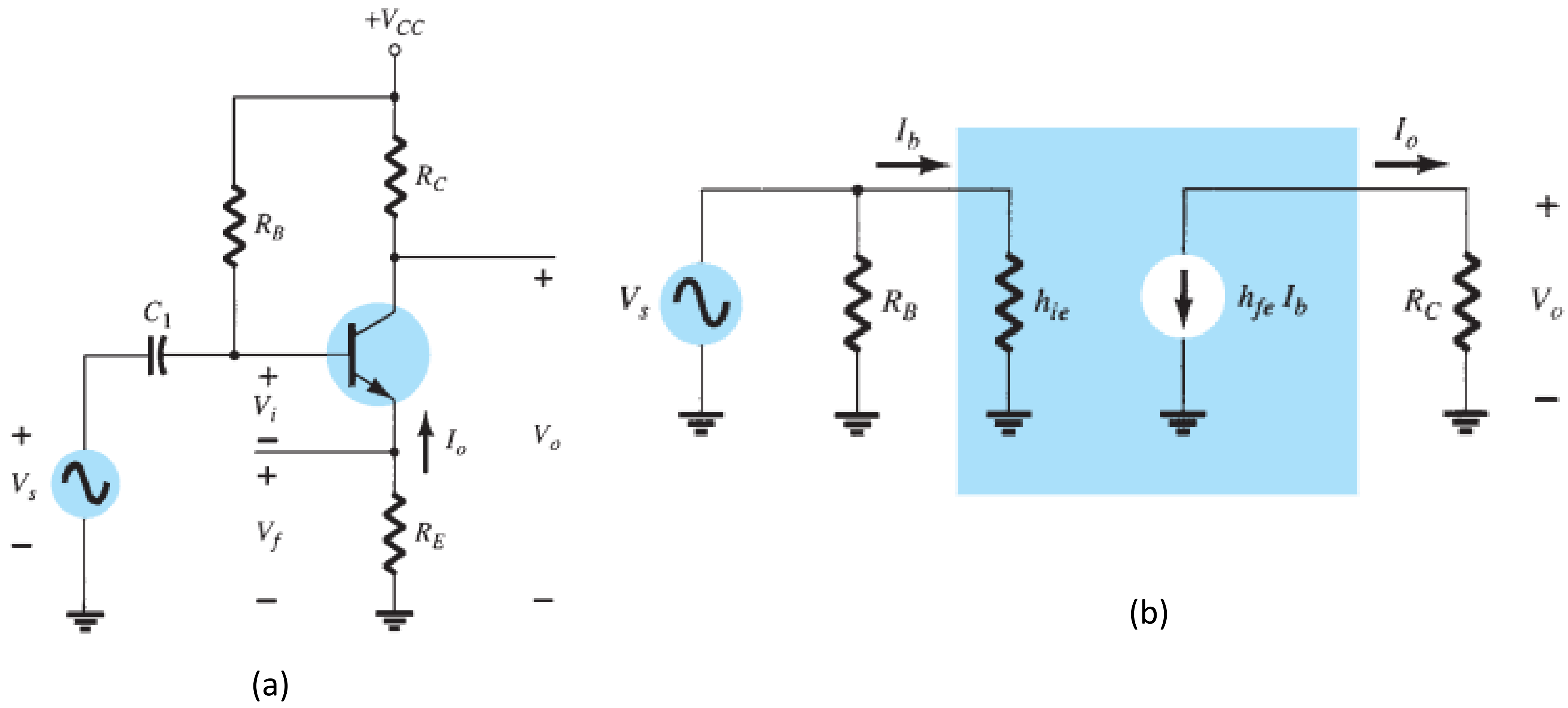
$$A_f \cong 1$$



Practical Feedback Circuits

Current-series feedback amplifier

Another feedback technique is to sample the output current Io and return a proportional voltage in series with the input. Although it stabilizes the amplifier gain, the current-series feedback connection increases input resistance. Figure shows a single transistor amplifier stage. Since the emitter of this stage has an unbypassed emitter, it effectively has current-series feedback. The current through resistor R E results in a feedback voltage that opposes the source signal applied, so that the output voltage V o is reduced. To remove the current-series feedback, the emitter resistor must be either removed or bypassed by a capacitor (as is usually done).



Transistor amplifier with unbypassed emitter resistor (R E) for current-series feedback: (a) amplifier circuit; (b) ac equivalent circuit without feedback

Without feedback

$$A = \frac{I_o}{V_i} = \frac{-I_b h_{fe}}{I_b h_{ie} + R_E} = \frac{-h_{fe}}{h_{ie} + R_E}$$

$$\beta = \frac{V_f}{I_o} = \frac{-I_o R_E}{I_o} = -R_E$$

The input and output impedances are, respectively,

$$Z_i = R_B \| (h_{ie} + R_E) \cong h_{ie} + R_E$$

$$Z_o = R_C$$

With feedback

$$A_f = \frac{I_o}{V_s} = \frac{A}{1 + \beta A} = \frac{-h_{fe}/h_{ie}}{1 + (-R_E)\left(\dfrac{-h_{fe}}{h_{ie} + R_E}\right)} \cong \frac{-h_{fe}}{h_{ie} + h_{fe} R_E}$$

The input and output impedances are calculated as specified in Table

$$Z_{if} = Z_i(1 + \beta A) \cong h_{ie}\left(1 + \frac{h_{fe} R_E}{h_{ie}}\right) = h_{ie} + h_{fe} R_E$$

$$Z_{of} = Z_o(1 + \beta A) = R_C\left(1 + \frac{h_{fe} R_E}{h_{ie}}\right)$$

The voltage gain A with feedback is

$$A_{vf} = \frac{V_o}{V_s} = \frac{I_o R_C}{V_s} = \left(\frac{I_o}{V_s}\right) R_C = A_f R_C \cong \frac{-h_{fe} R_C}{h_{ie} + h_{fe} R_E}$$